

Integration of an SiPM-Based Scintillation Detector into the Phase-2 CMS Drift Tube Readout Chain Using the IpGBT ASIC

Master's thesis in physics

Nils Jonathan Esper

Supervisor: Univ.-Prof. Dr. rer. nat. Alexander Schmidt

Second reviewer: Apl. Prof. Dr. rer. nat. Oliver Pooth



Physics Institute III A, RWTH Aachen University

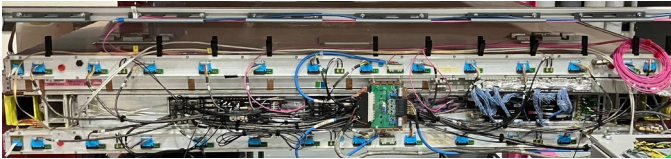
Aachen, 19th December 2025

Introduction

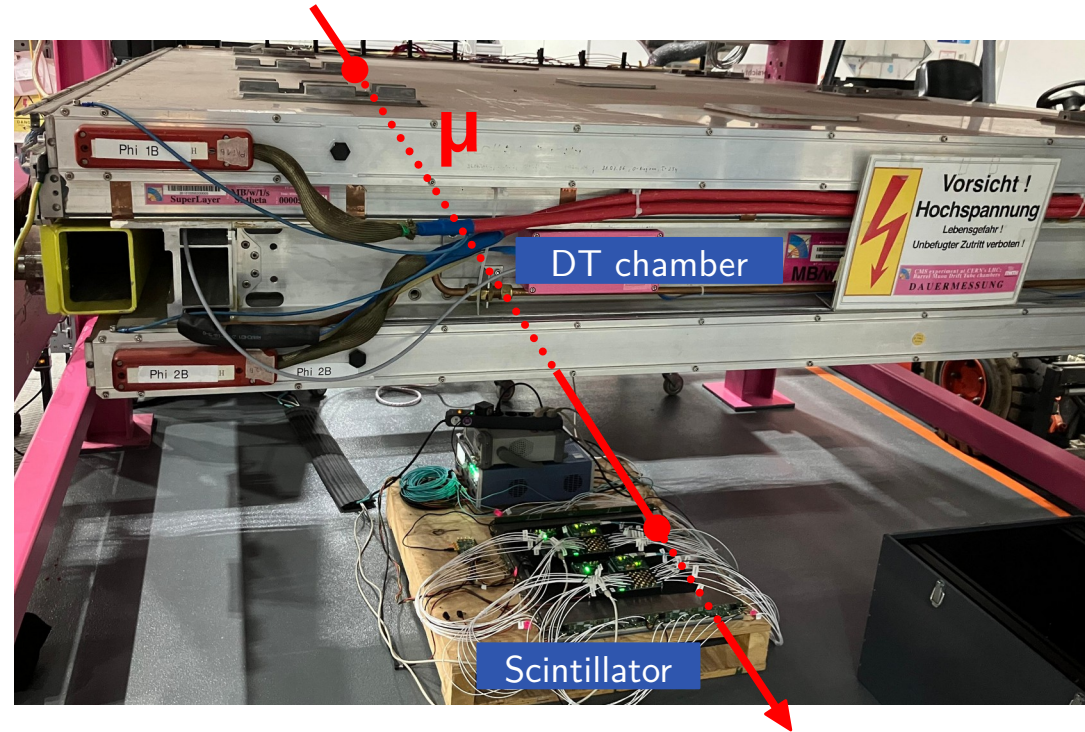
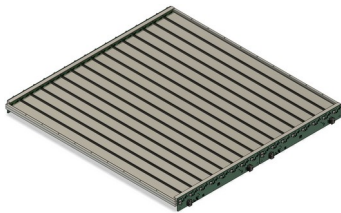
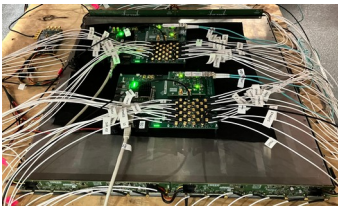
Introduction

Two different muon detectors available:

- CMS Drift Tube (DT) chamber:



- Scintillator strip detector with SiPM readout:



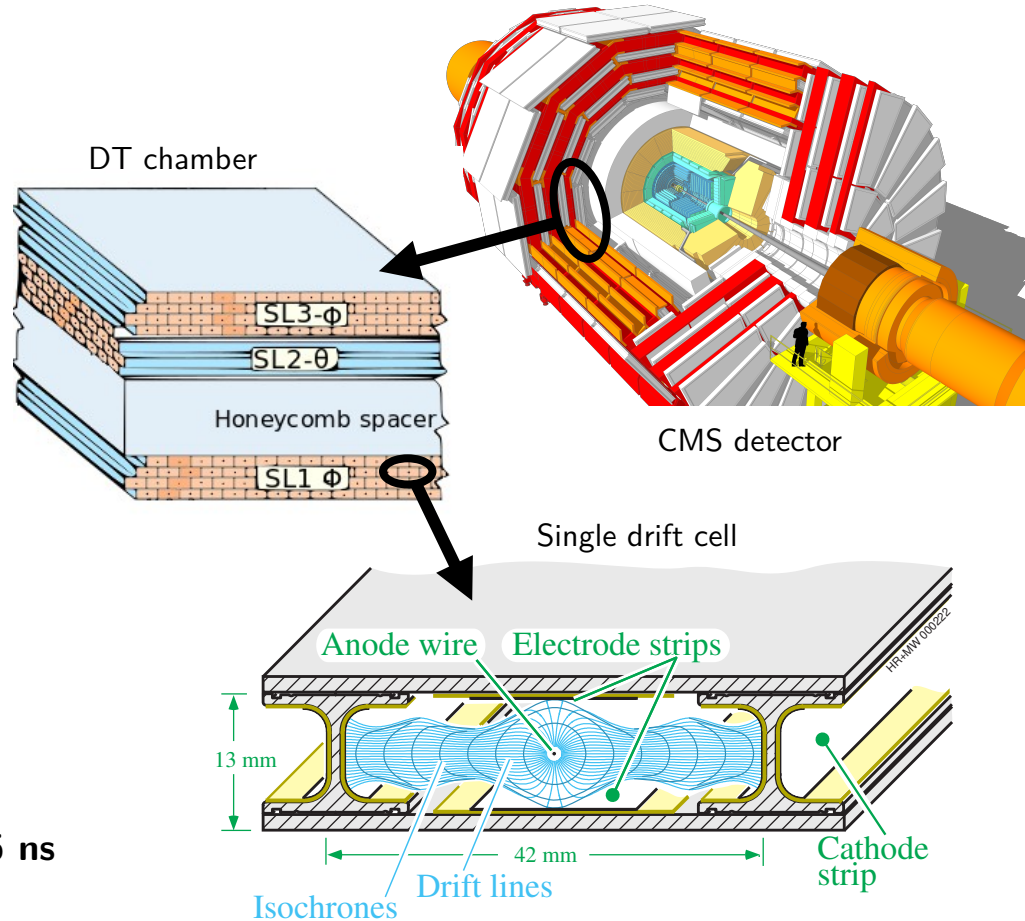
Idea for this thesis:

- **Integrate scintillator into the CMS DT readout chain at cosmic test stand**
 - **Demonstrate synchronization of detectors with atmospheric muon data**

CMS Drift Tube chamber

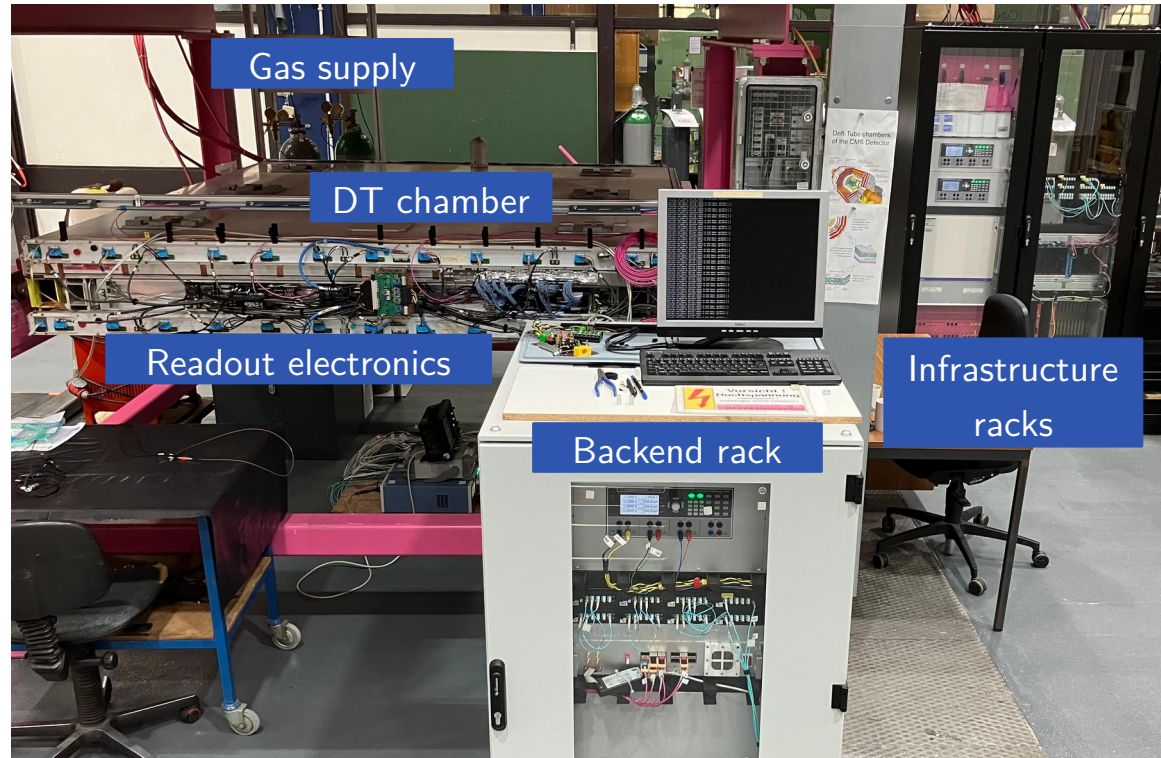
CMS DT system

- Compact Muon Solenoid (CMS) detector:
 - Multipurpose particle detector at Large Hadron Collider (LHC) at CERN
- Drift Tube (DT) chambers:
 - Gaseous detector in CMS barrel region $|\eta| < 1.2$
 - **3 superlayers** with **4 layers** of $O(50)$ **drift cells** each
 - SL 2 rotated relative to SLs 1, 3
- Drift cell:
 - Cross section: **42 x 13 mm²**
 - Length: $O(2 \text{ m})$
 - Gas mixture: 85% Ar, 15% CO₂
 - Electric field: $O(1.5 \text{ kV/cm})$
 - Ionization electrons drift with $\mathbf{v_{drift} = 54.5 \text{ } \mu\text{m/ns}}$
 - Cell half-width determines max. drift time: $\mathbf{t_{drift,max} = 385 \text{ ns}}$



Cosmic test stand in Aachen

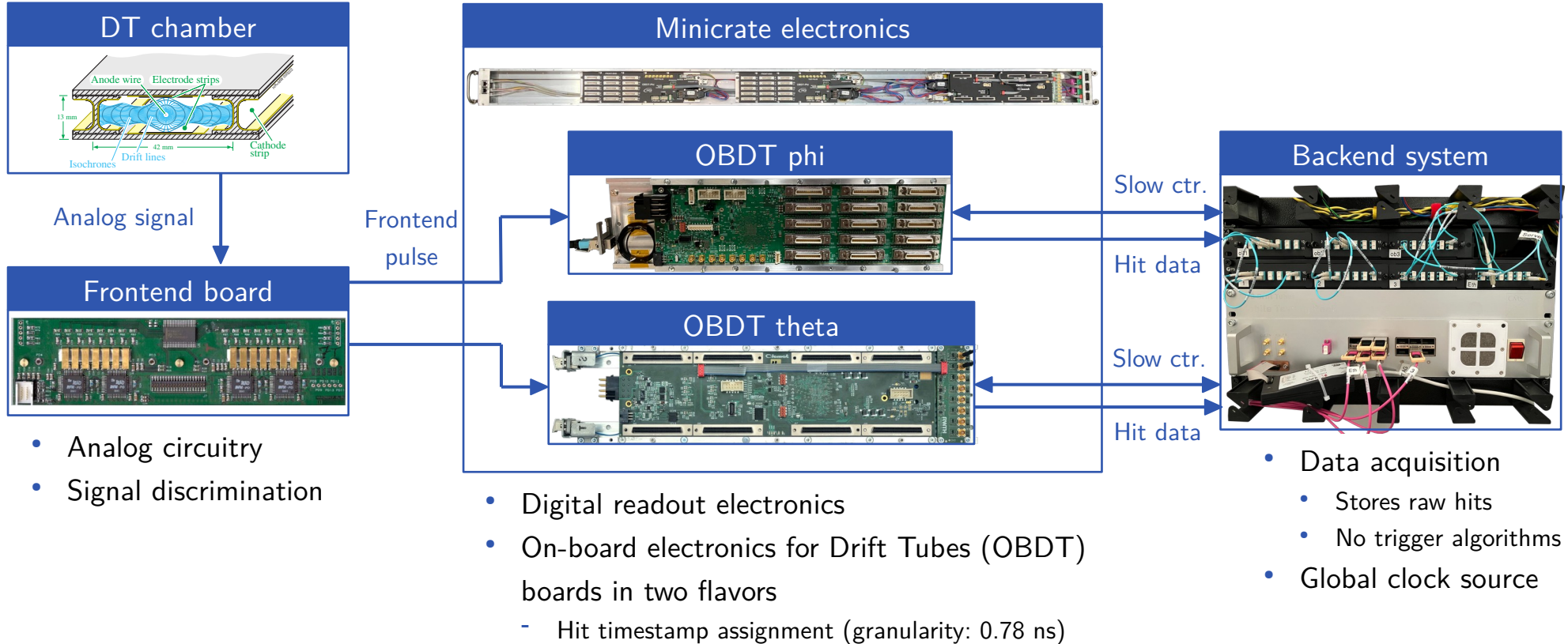
- Our institute operates cosmic test stand in experimental hall
 - 25% of DT chambers for CMS produced in Aachen in early 2000s
 - Spare chamber left at test stand
- Features fully functional CMS DT chamber
 - Dimensions: $2.5 \times 2.2 \times 0.4 \text{ m}^3$



Setup at cosmic test stand

DT readout electronics

- Resembles CMS DT Phase-2 readout chain



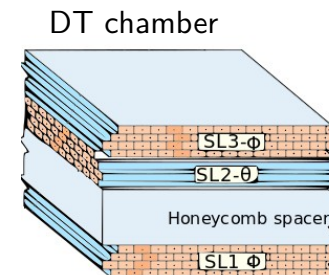
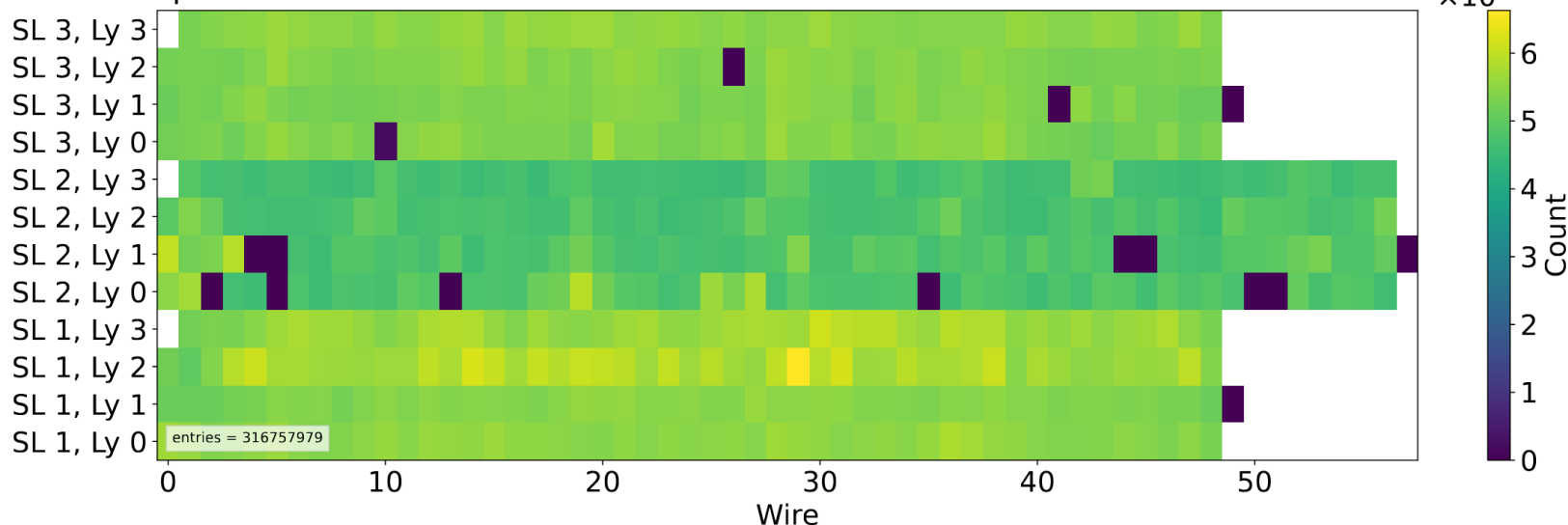
Recorded Drift Tube chamber data

DT hit data

- Recorded large dataset of atmospheric muons with DT chamber:

Duration	DT hits
5h 27min	316,757,979

Wire occupancies:



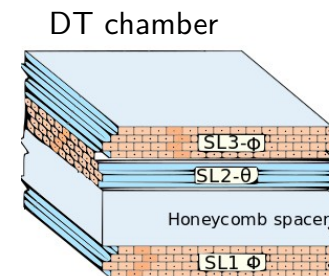
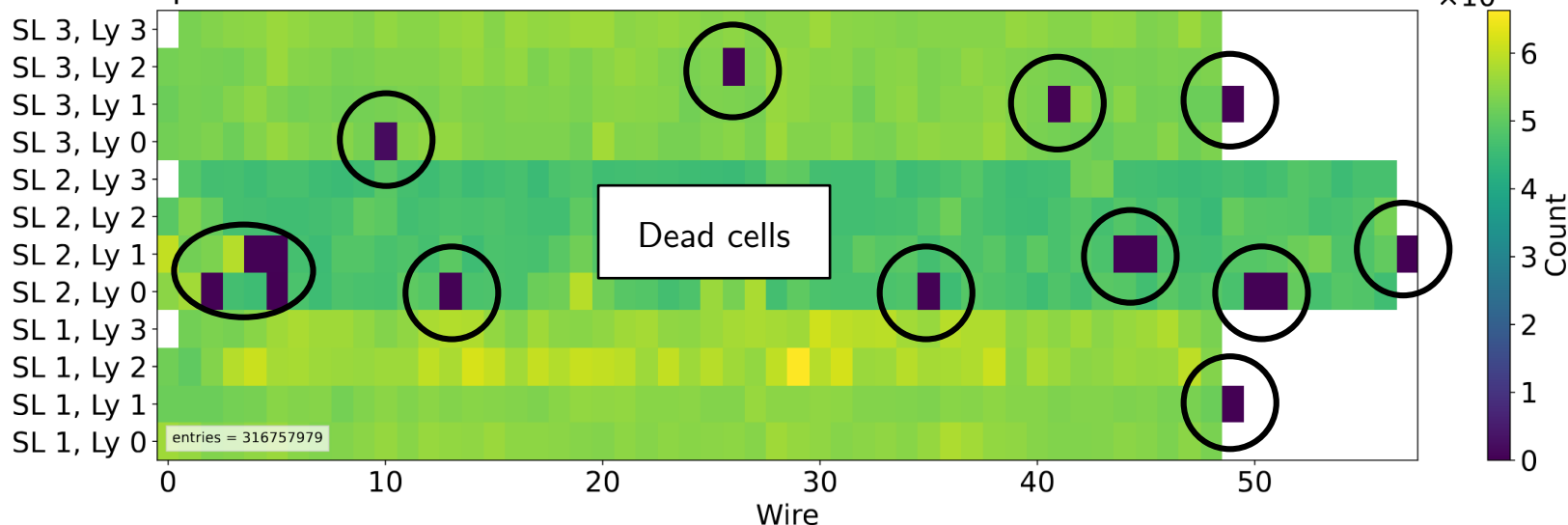
- Observe uniform wire rates: **28.0 Hz** (SLs 1, 3) and **24.5 Hz** (SL 2)
 - Rate differences because of different wire lengths (chamber size: $2.5 \times 2.2 \text{ m}^2$)
- 16 dead cells** in the chamber

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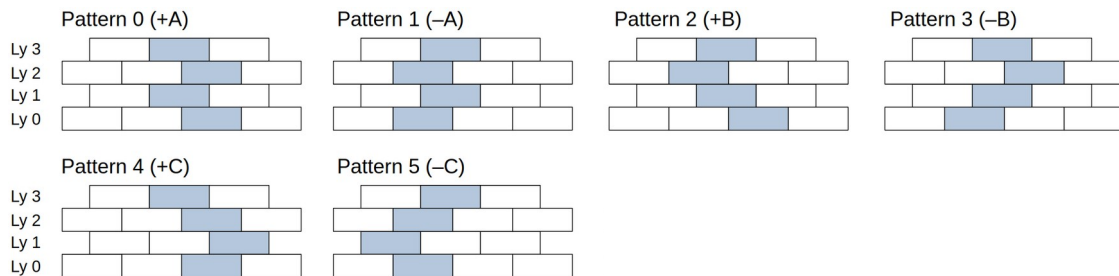
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DT superlayer patterns

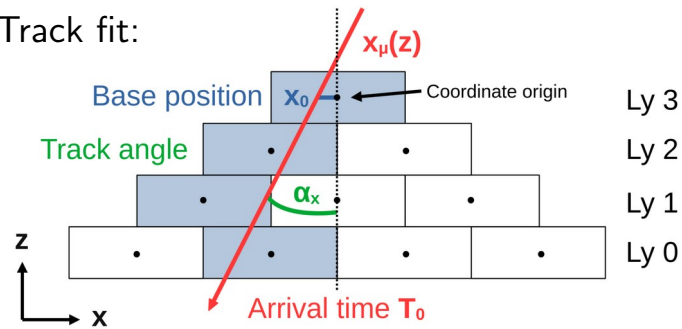
Triggerless track reconstruction:

- ▶ Treat all cell hits as potential track hits
- Superlayer pattern:
 - Exclude hits that could not be created simultaneously: $\Delta T_{\max} \leq t_{\text{drift},\max}$
 - Pattern should have **track-like shape**
 - Require 4 hits in pattern
- Superlayer pattern fit:
 - **Fit local track** to pattern hits
 - Relate hit timestamps to linear track
 - Assume horizontal drift with $v_{\text{drift}} = \text{const.}$
 - **Local track parameters:** T_0 , x_0 , $\tan \alpha$
 - **Apply cuts** to reduce noise patterns
 - Goodness-of-fit variable $\chi^2/N_{\text{df}} < 10$
 - Remove unphysical drift times

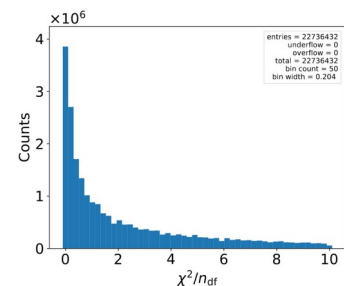
Superlayer pattern geometry:



Track fit:



χ^2/N_{df} distribution:

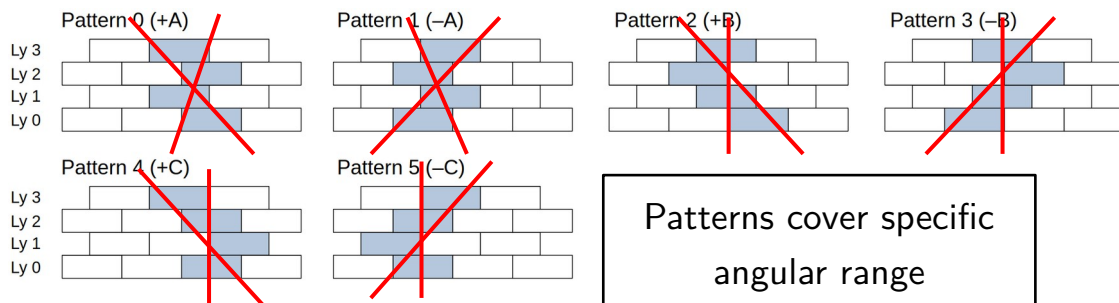


DT superlayer patterns

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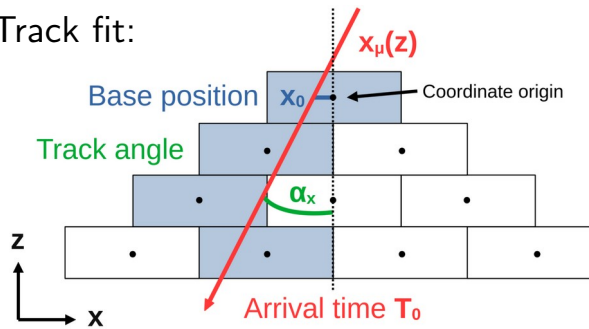
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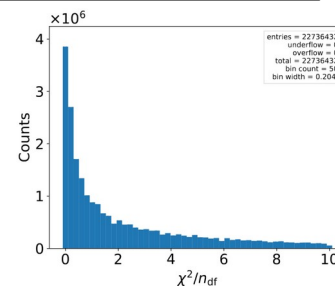


Patterns cover specific angular range
→ Reconstruction only up to max. angle

Track fit:

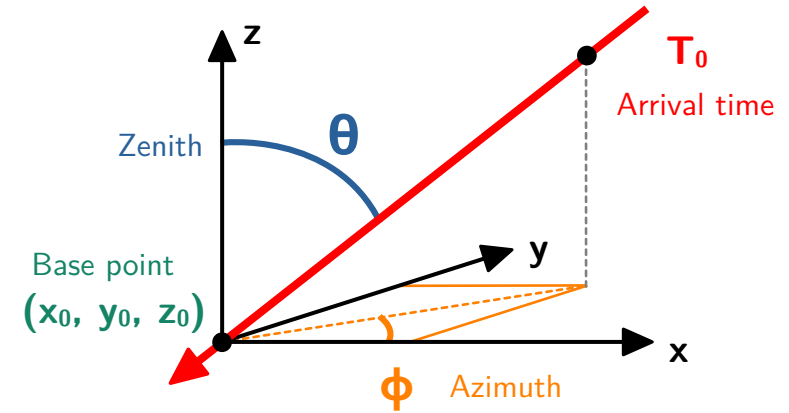


Ly 3
Ly 2
Ly 1
Ly 0



DT global track building

- Temporal matching of pattern fits:
 - Require fitted pattern in all 3 superlayers
 - Select patterns close in time: $\Delta T_{0,SL} \leq 20 \text{ ns}$
- Translation of track projections into global frame:
 - **Global track parameters:** T_0 , (x_0, y_0, z_0) , ϕ , θ



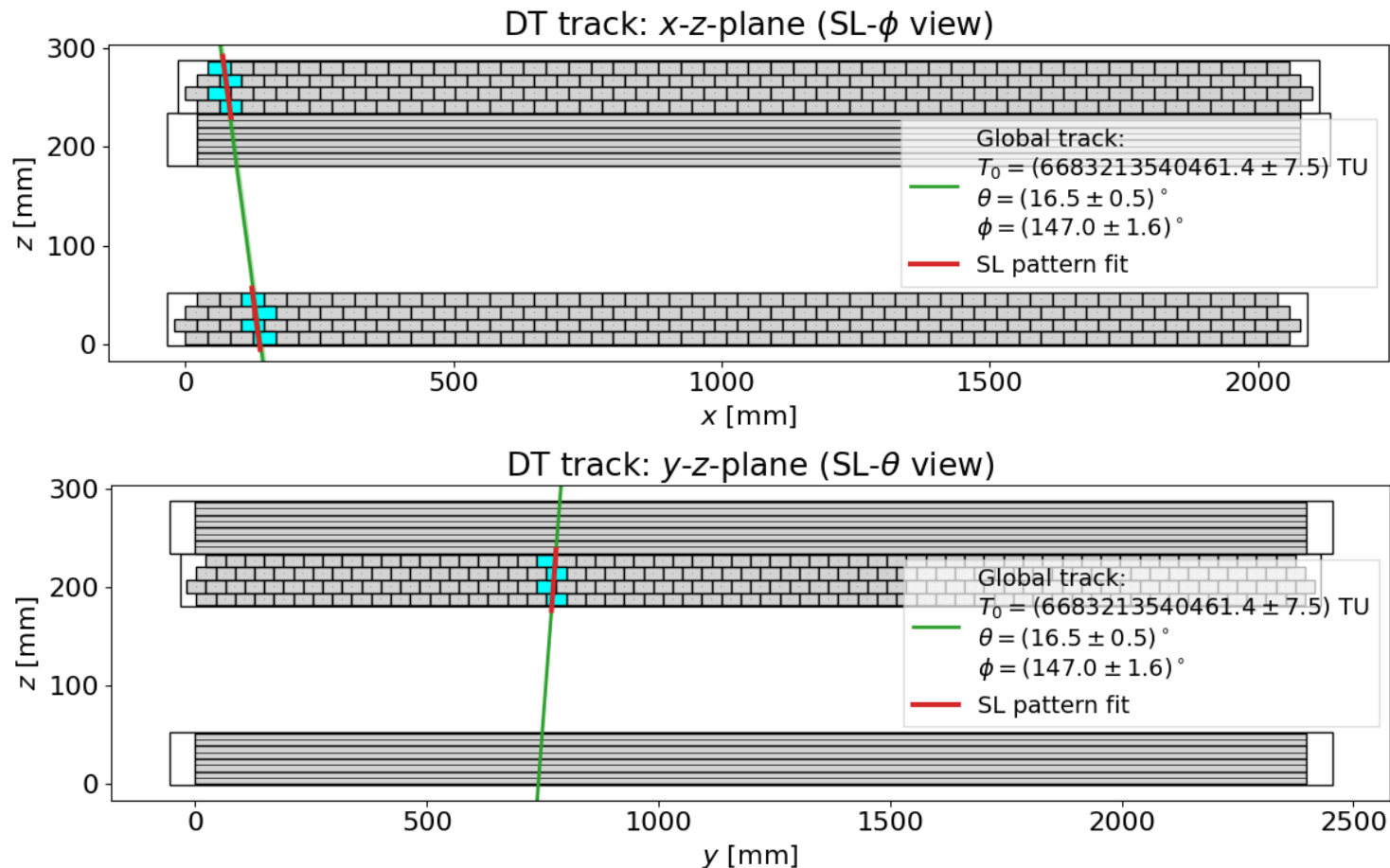
- ▶ Reconstructed global muon rate: **85 Hz**
 - ▶ Global track parameters (mostly) behave as expected

Simplified atmospheric muon simulation: **196 Hz**

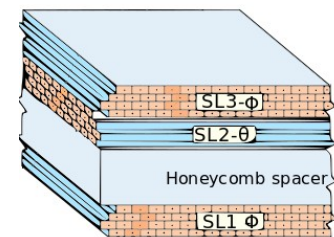
- ▶ Should be understood as upper limit
 - ▶ Simulated hits are generated according to fit assumptions
 - ▶ No modeling of noise or (drift time) measurement uncertainties

DT global track results

Example reconstructed track, lateral view:

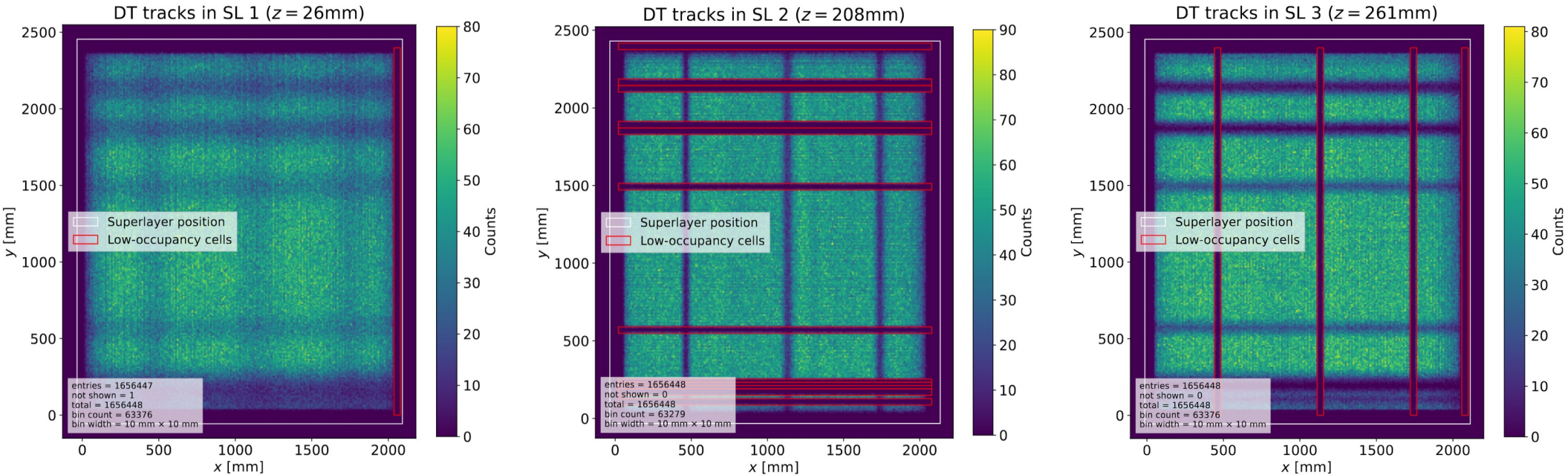


DT chamber



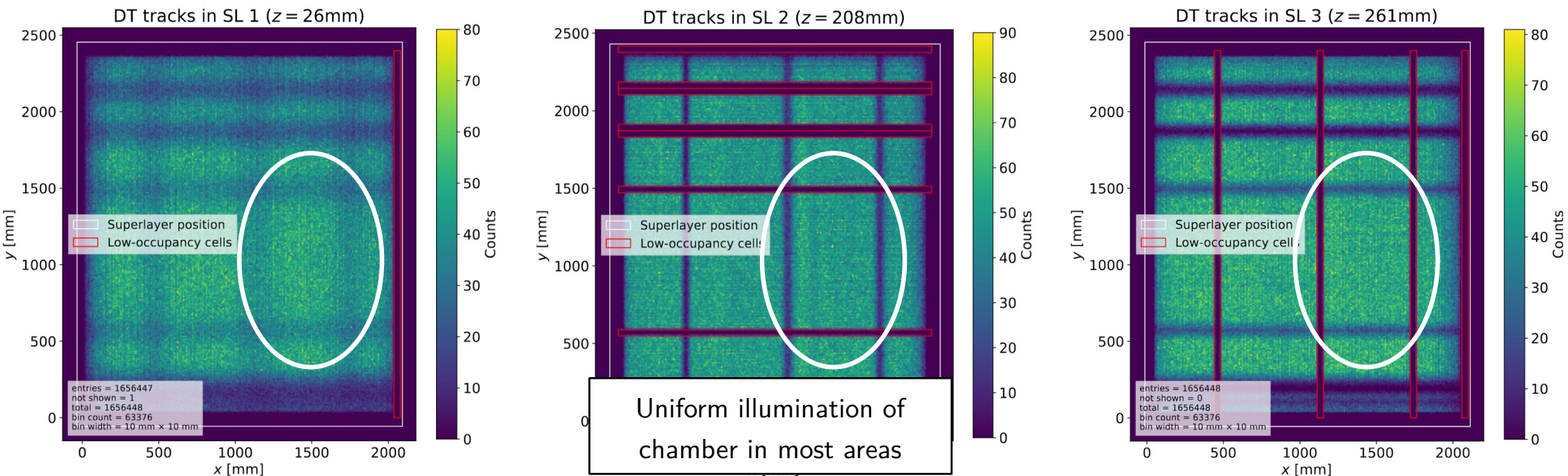
DT global track results

2D histograms of global track positions in superlayers, top view:



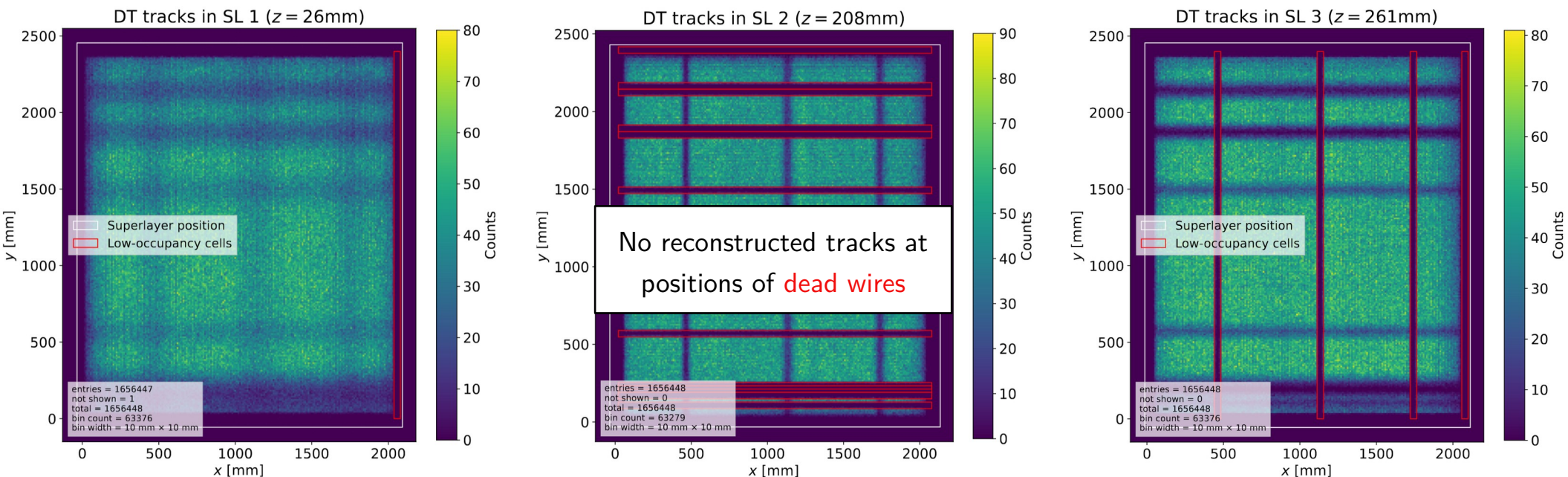
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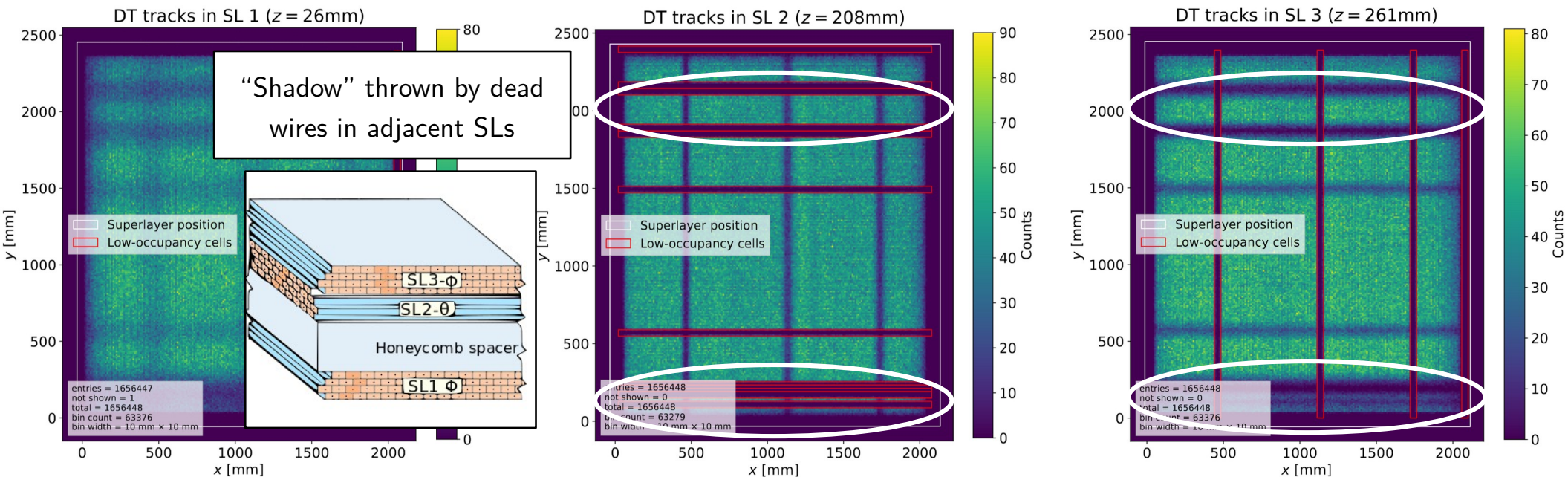
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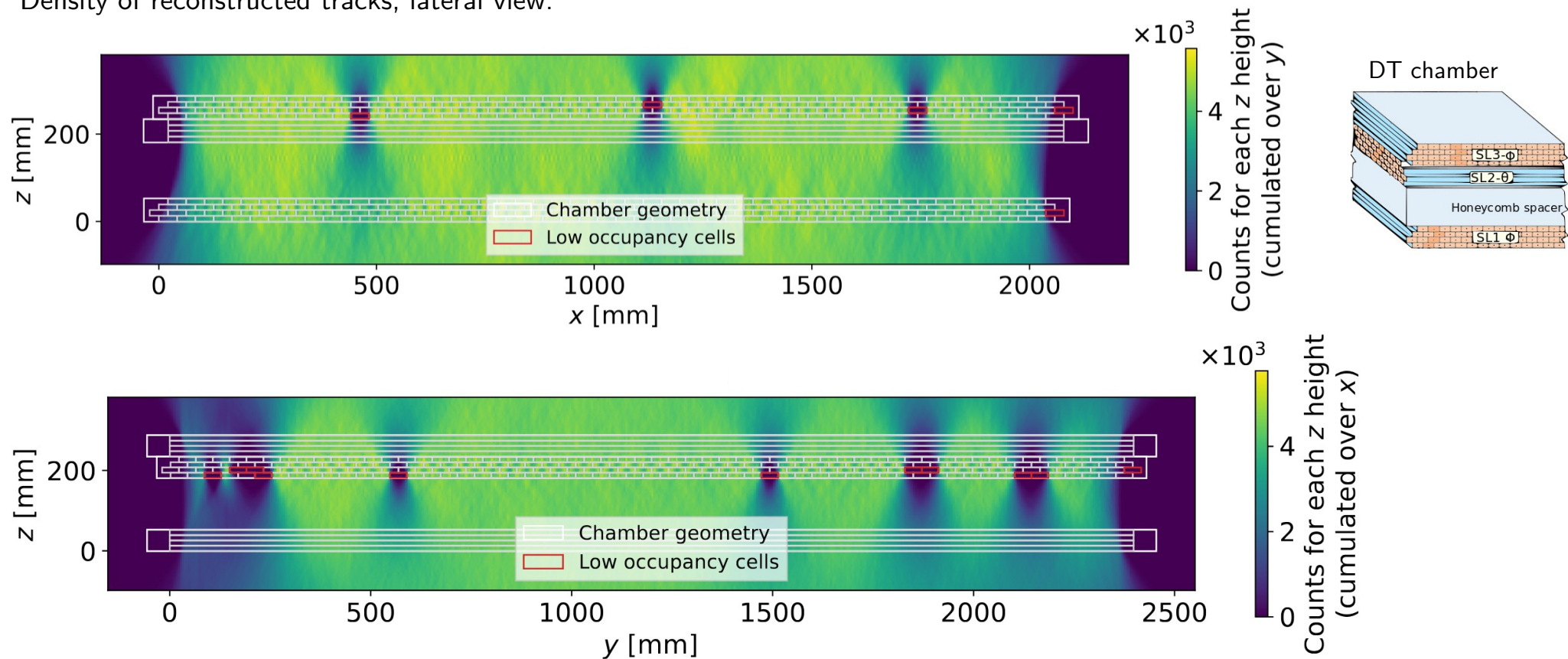
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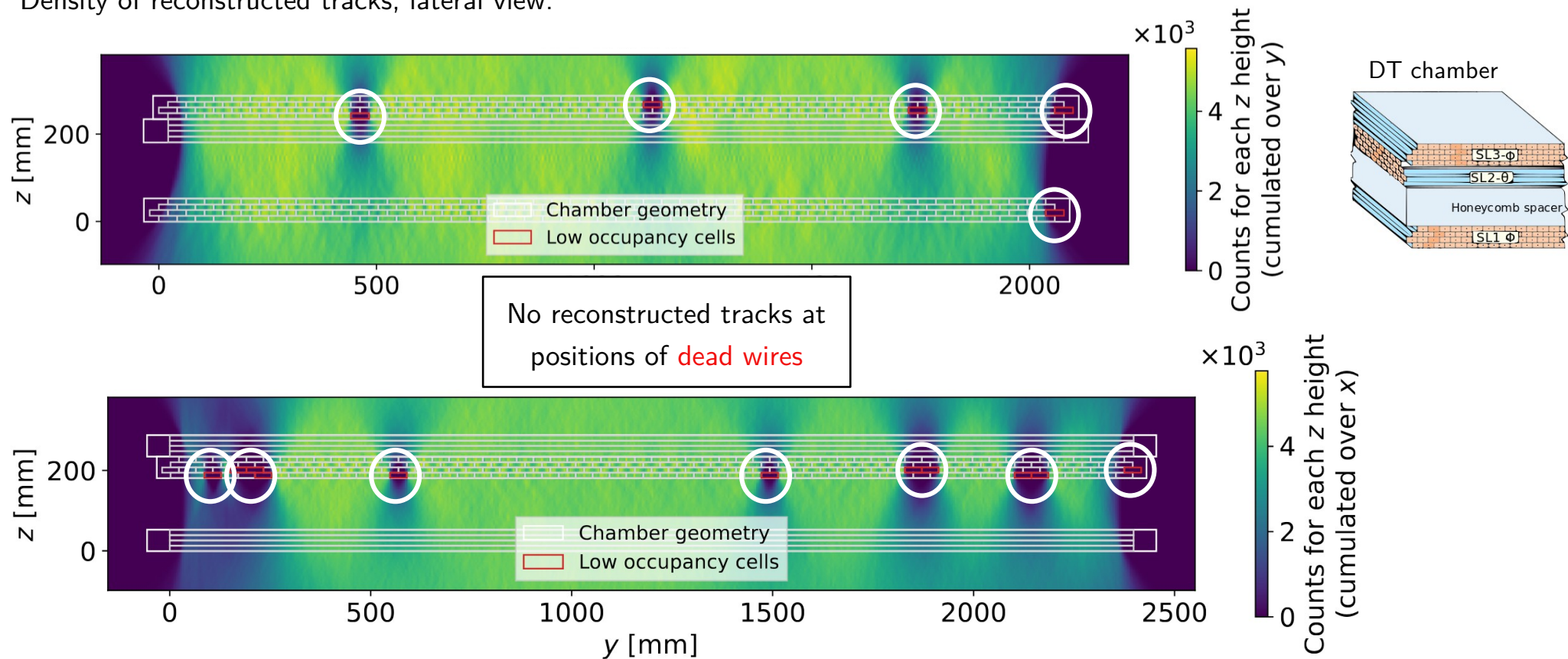
DT global track results

Density of reconstructed tracks, lateral view:



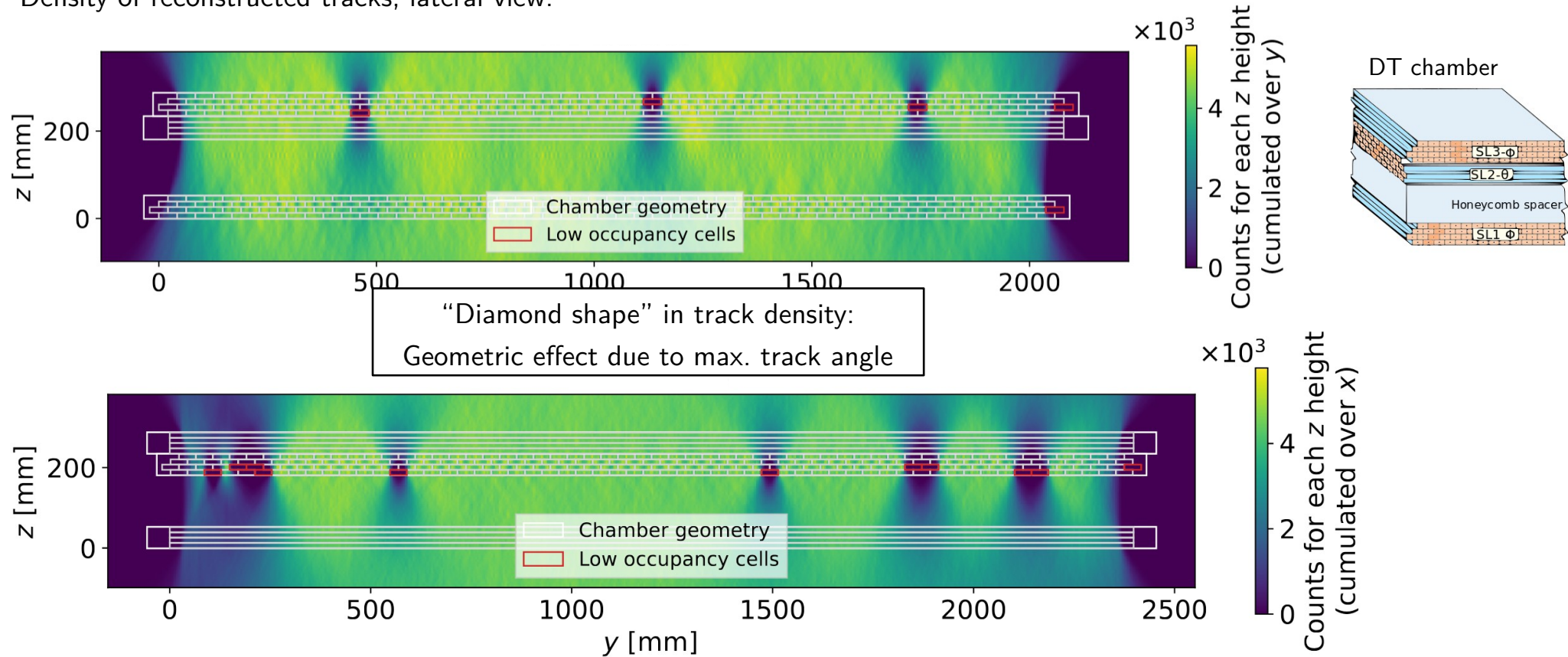
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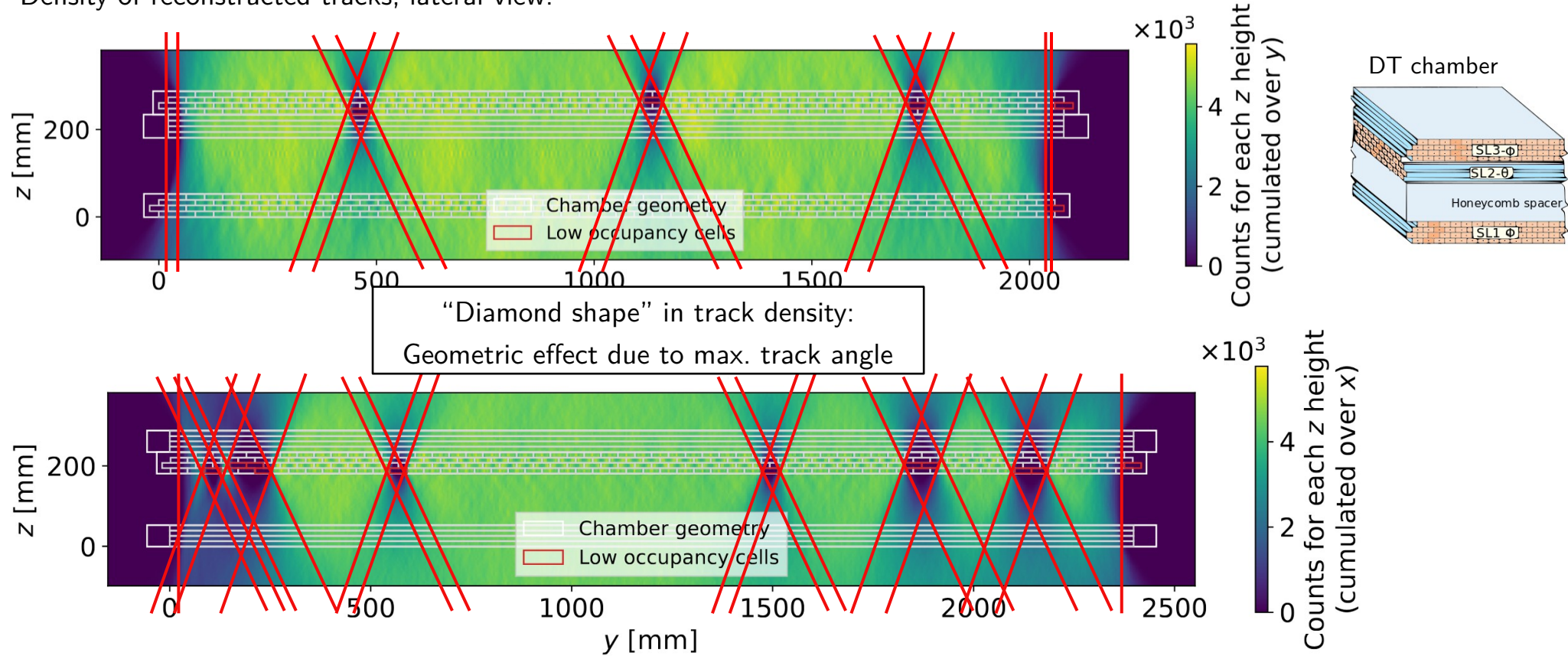
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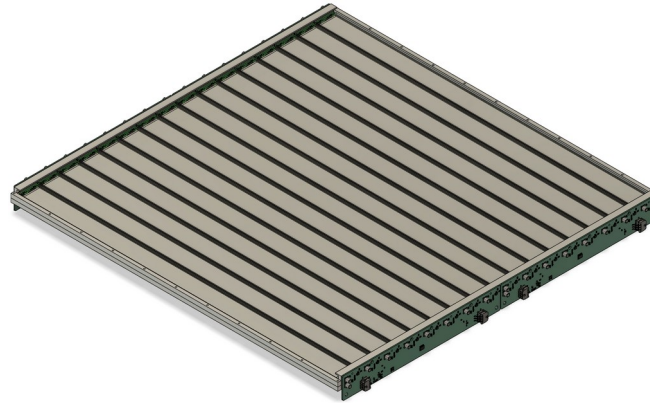
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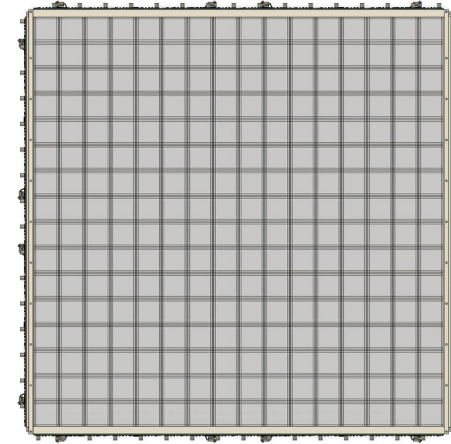
Scintillator strip detector

Scintillator strip detector with SiPM readout

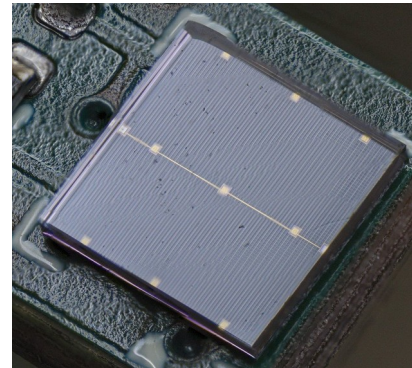
- Scintillator strip detector:
 - Developed in previous thesis projects (E. Ehlert)
 - Plastic scintillator material (EJ-200)
 - Detector size: $50 \times 50 \text{ cm}^2$
 - Strip width: 3 cm
 - 2 layers with 16 strips each
 - Photon readout: 2 SiPMs / strip
 - 64 SiPMs in total
- Silicon photomultiplier (SiPM):
 - Solid-state detector with single-photon sensitivity
 - Array of Geiger Avalanche Photodiodes (GAPDs)
 - Single cell hit: 1 p.e. (photoelectron equivalent)
 - Gain depends on applied bias voltage



Scintillator strips



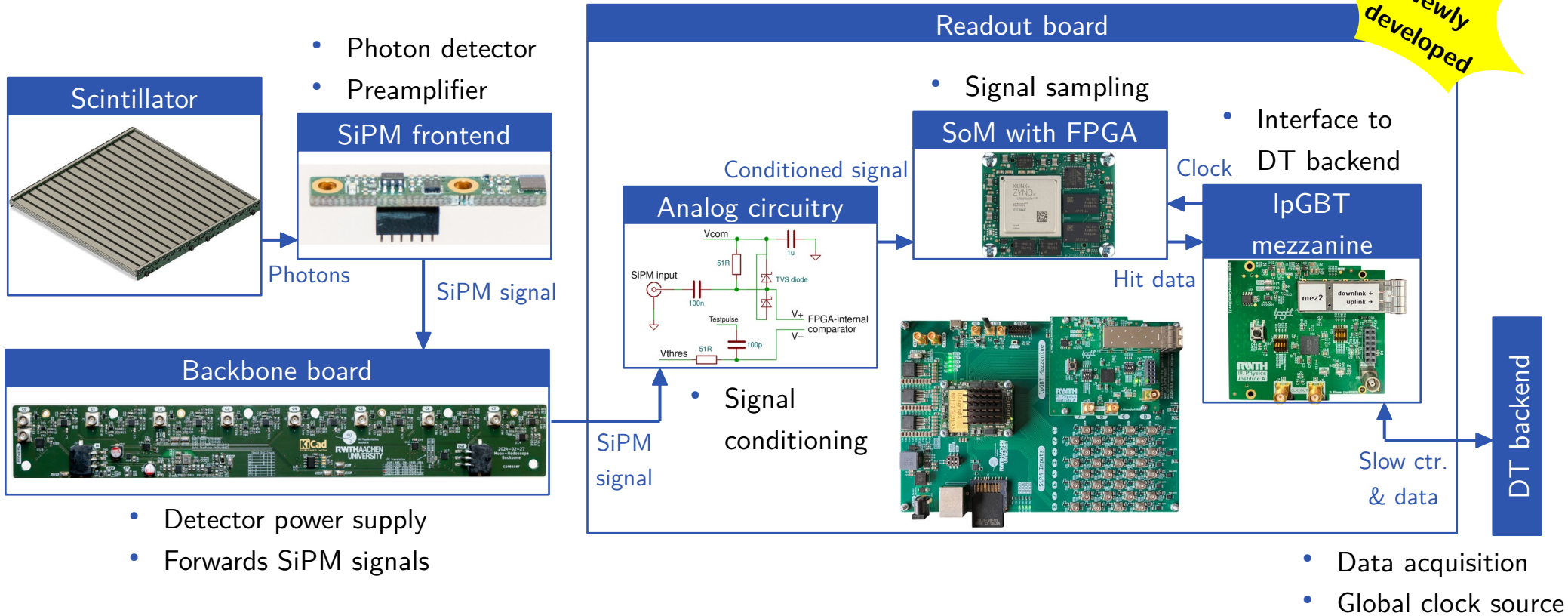
Scintillator top view



Mounted SiPM

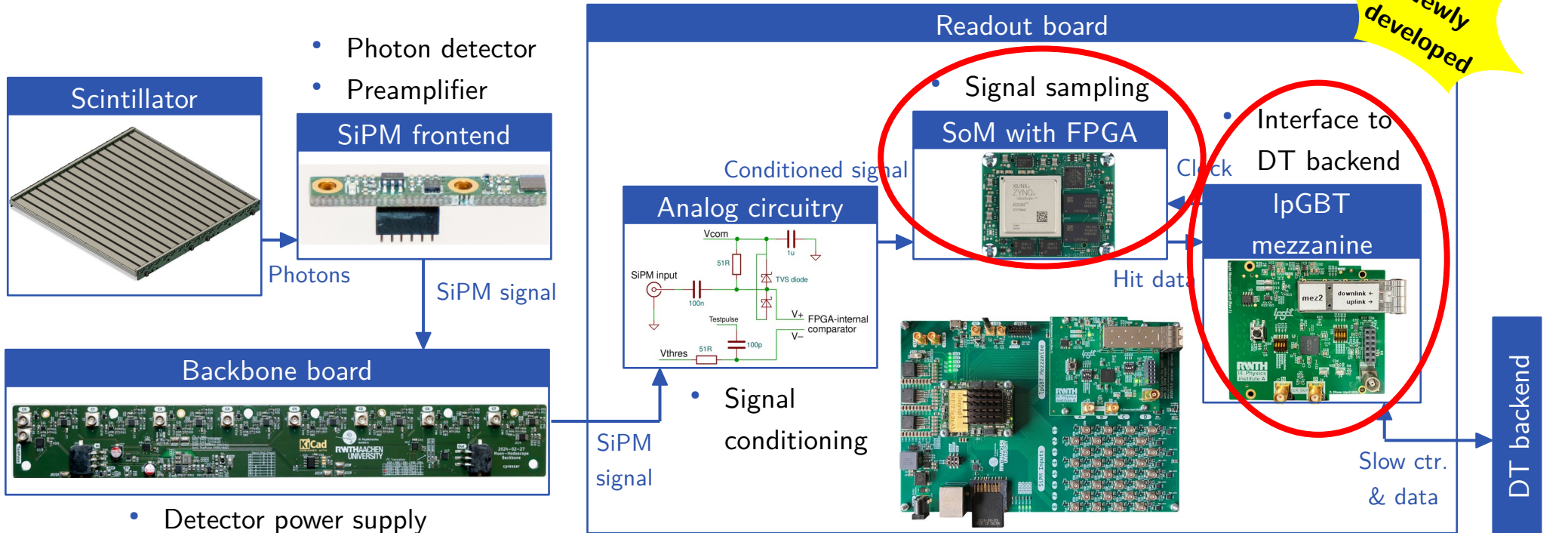
Scintillator readout electronics – Overview

- ▶ Should have capability to interface DT backend



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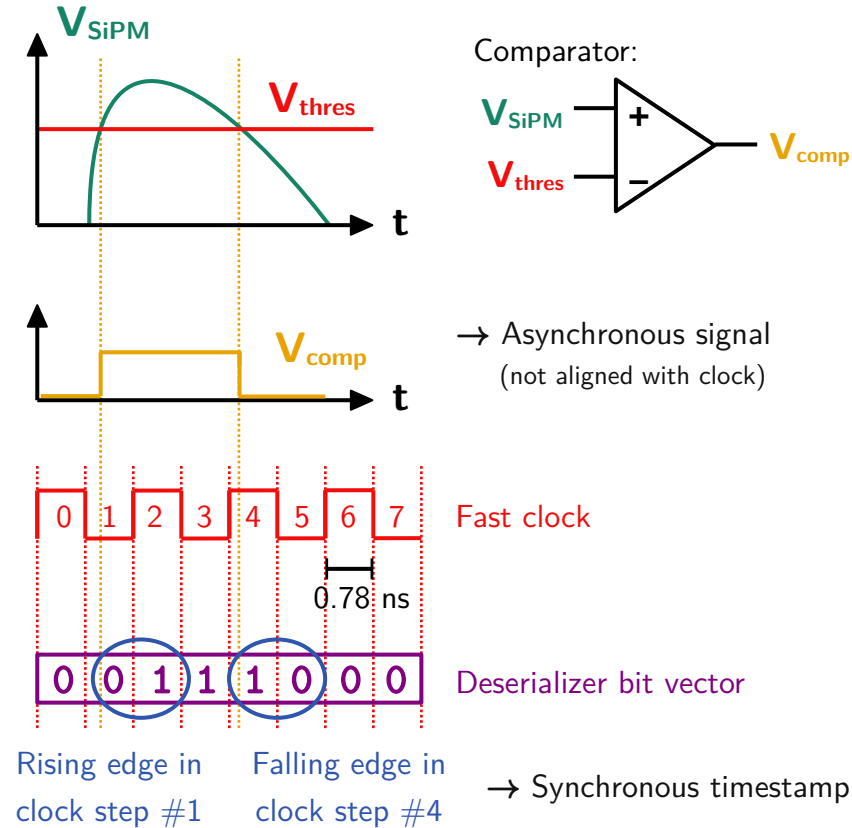
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- Data acquisition
- Global clock source

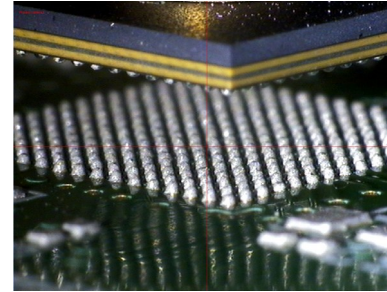
Scintillator readout electronics – Signal sampling

- Receive preamplified analog signal from SiPM
- Discriminator:
 - Hit discrimination threshold: $V_{\text{thres}} = 25 \text{ mV}$ (~ 5 p.e. of SiPM)
 - Implemented with FPGA-internal comparators
- Time-to-Digital Converter (TDC):
 - Assignment of digital timestamp
 - Implementation in FPGA firmware (using deserializers)
 - Deserializer returns bit vector of comparator values, sampled at each clock edge
 - Apply edge detection logic to bit vector

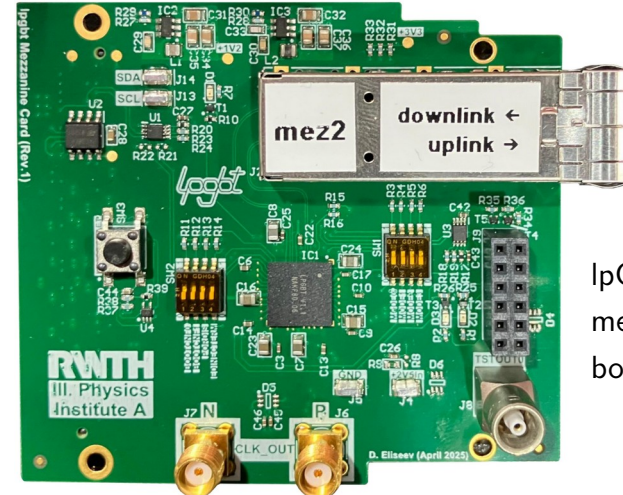


Scintillator readout electronics – lpGBT mezzanine

- Low-Power Gigabit Transceiver (lpGBT):
 - Chip for high-speed bidirectional data transfer
 - Extensively used in Phase-2 Upgrade of CMS
 - Capability to recover synchronous clock from backend
- lpGBT mezzanine:
 - Self-developed board which hosts lpGBT chip
 - FPGA interface: Serial data lines (eLinks)
 - DT backend interface: Optical fiber



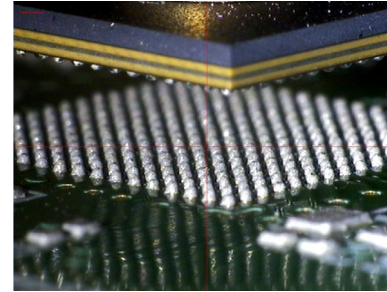
lpGBT chip being soldered onto a board



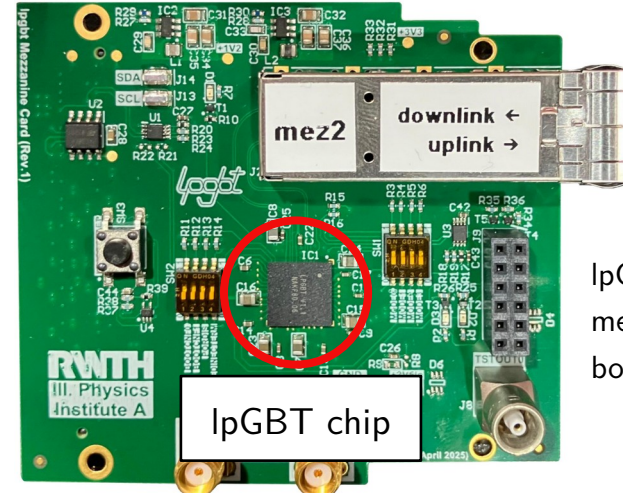
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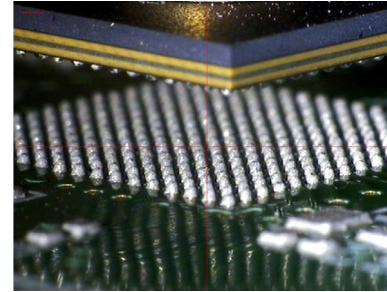
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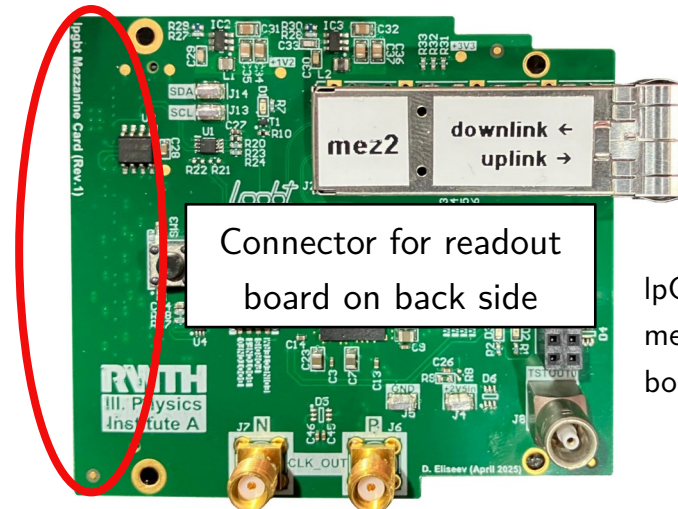
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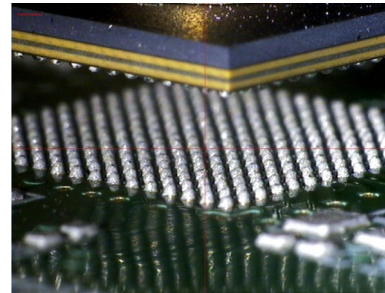
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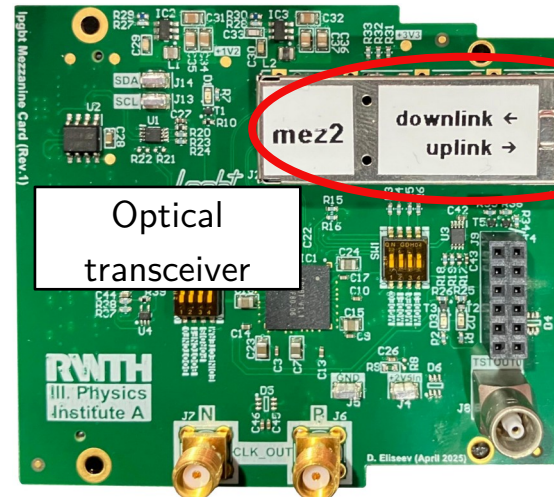
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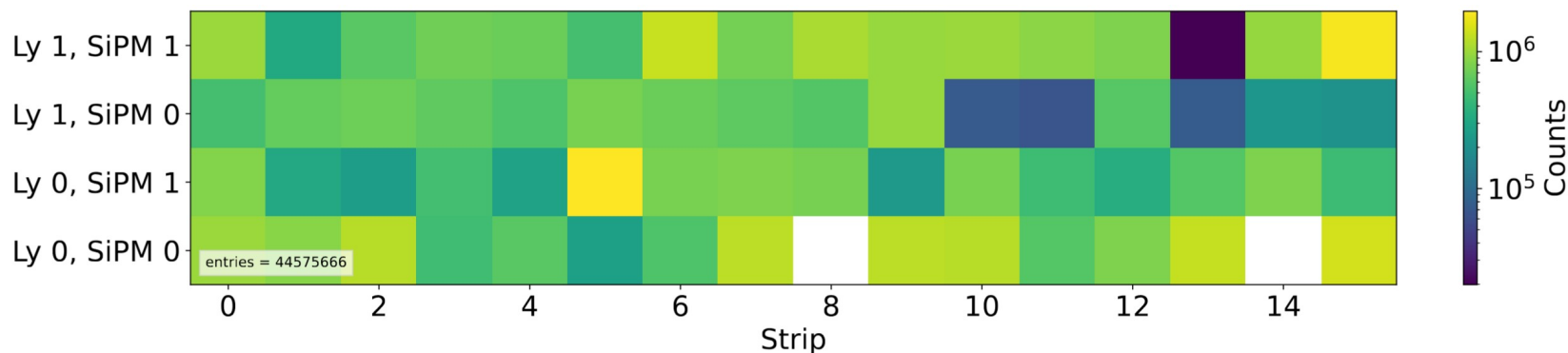
Recorded scintillator strip detector data

SiPM hit data

- Recorded large dataset of atmospheric muons with scintillator:
 - Suppress systematic readout effects: Apply dead time of 250 ns

Duration	SiPM hits
5h 27min	44,575,666

SiPM occupancies:



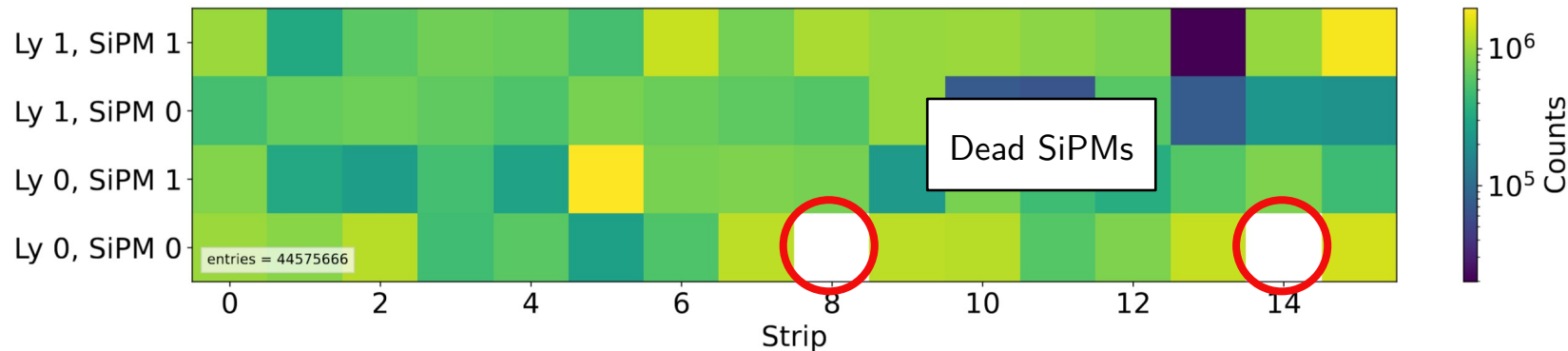
- Average SiPM rate: **37 Hz**
 - Rate fluctuations over one order of magnitude
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- 2 dead SiPMs** in detector

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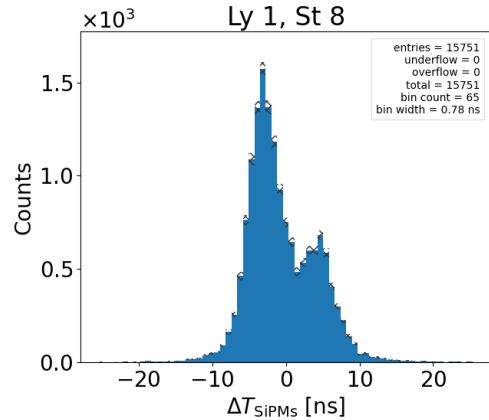


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Scintillator strip coincidences

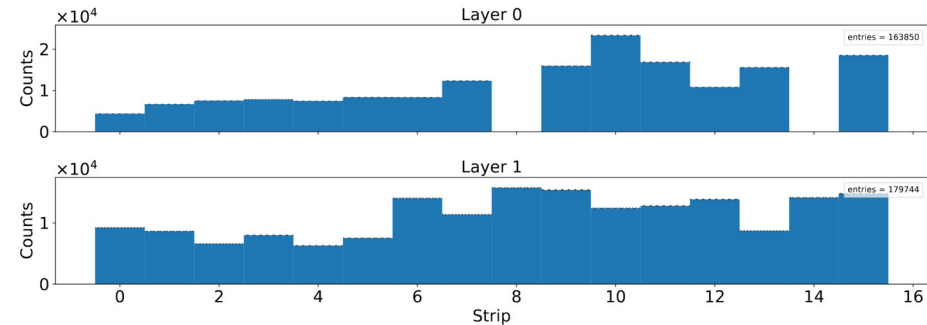
- Evaluate SiPM coincidences for scintillator strips
 - Coincidence window: $\Delta T_{\text{SiPMs}} \leq 25 \text{ ns}$
 - Require isolation from other hits in layer

Time difference ΔT_{SiPMs} of coincidence:



- ▶ **Contained coincidence timing $O(15 \text{ ns})$**
 - ▶ Indicates “real” coincidences
- ▶ Double-peak structure observed
 - ▶ Needs further investigation, possibly due to sampling logic

Obtained strip occupancies:



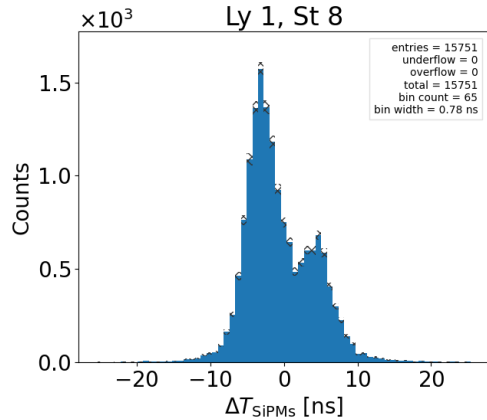
- ▶ Measured strip rate: **$O(0.5 \text{ Hz})$**
 - ▶ Observe strip rate fluctuations
 - SiPM bias voltage calibration should be performed
- ▶ **2 dead strips** from dead SiPM propagation

Simplified atmospheric muon simulation: **2.2 Hz**

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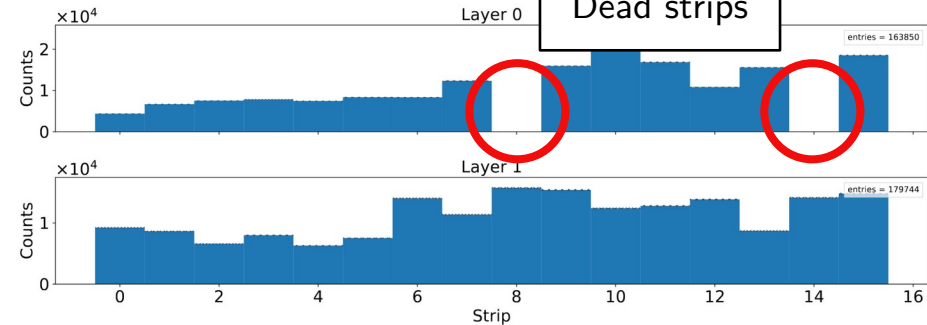
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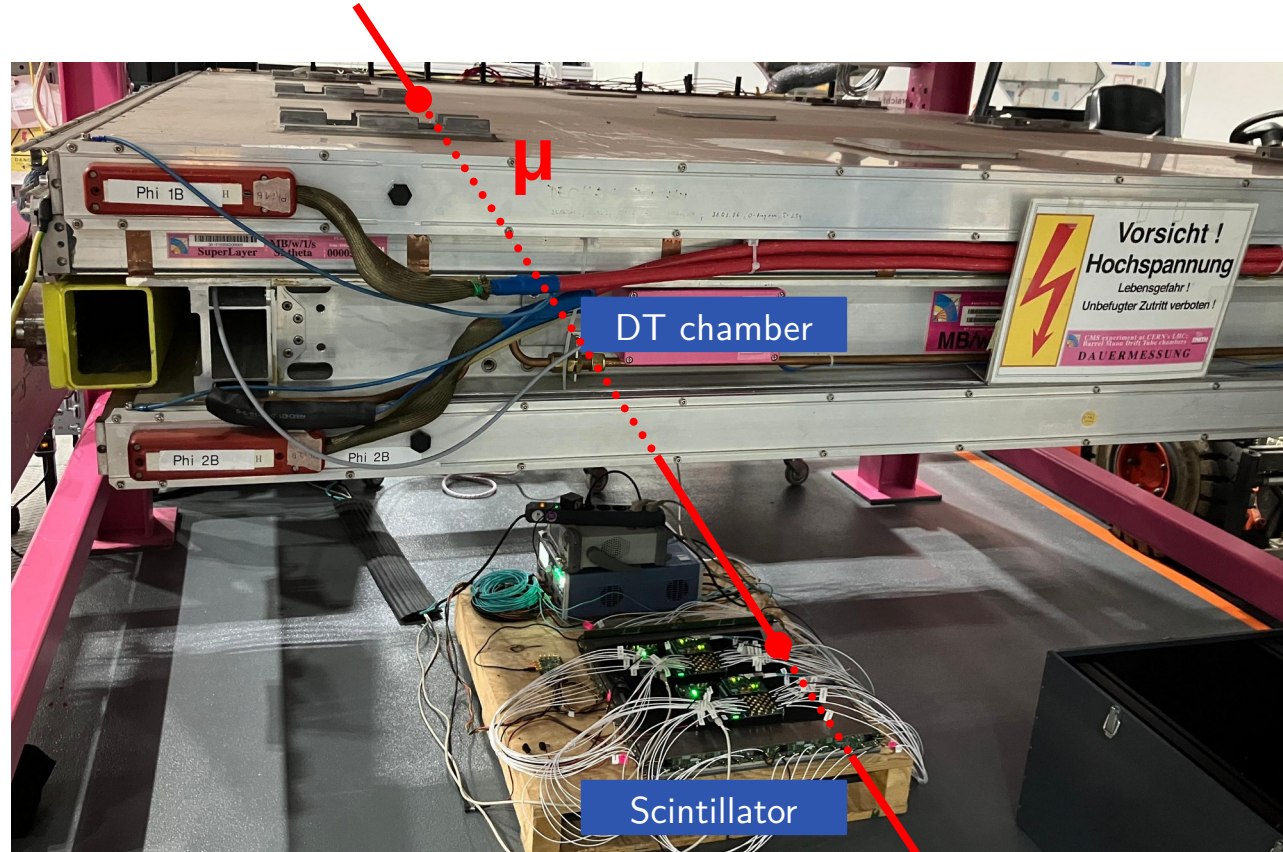
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Combination of detector data

Combined measurement setup

- Place scintillator below DT chamber at cosmic test stand
- ▶ **Attempt combined data analysis of reconstructed DT tracks & scintillator strip hits**



Temporal matching of DT tracks & scintillator hits

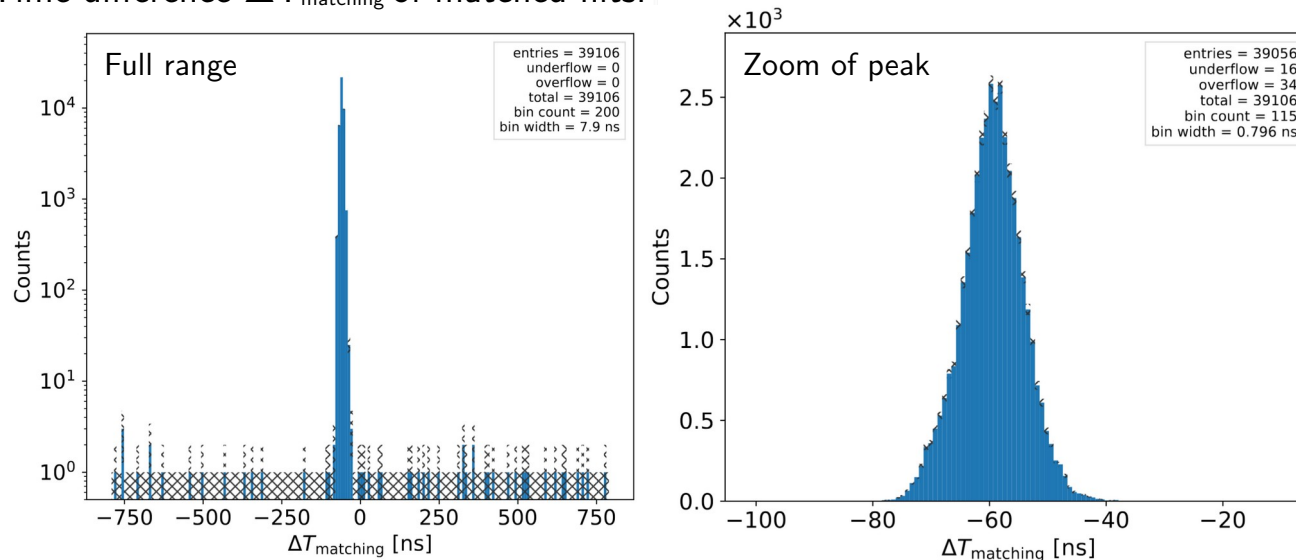
- Matching of detector hits by arrival time:
 - Require DT track & scintillator strip hit coincidence within 780 ns:
$$\Delta T_{\text{matching}} = T_{\text{scint}} - T_{\text{DT}}$$
 - Remove ambiguities while matching

- ▶ Matched hit rate: **2 Hz**

For comparison:

- Scaled DT track rate: 4 Hz
- Scintillator layer rate: 8 Hz

Time difference $\Delta T_{\text{matching}}$ of matched hits:



- ▶ **Successful detector synchronization !**

- ▶ **Peak FWHM: 12 ns**
- ▶ Includes all timing uncertainties of readout & reconstruction
- ▶ Observe constant **latency offset**
 - ▶ DT tracks are -59 ns later than scintillator hits
 - ▶ Possibly due to readout propagation difference

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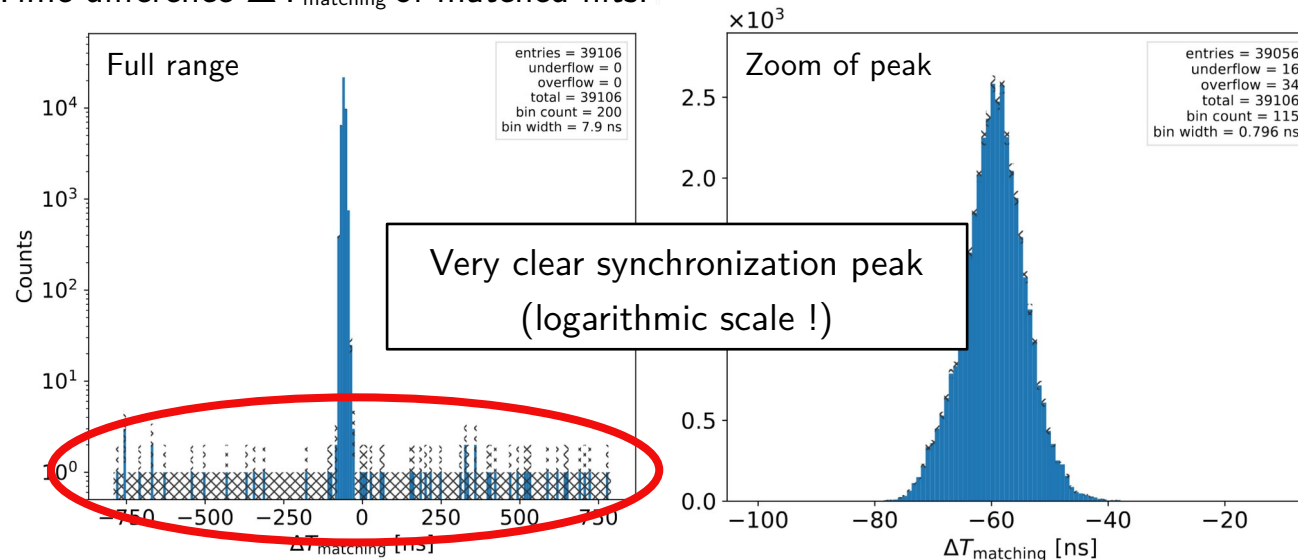
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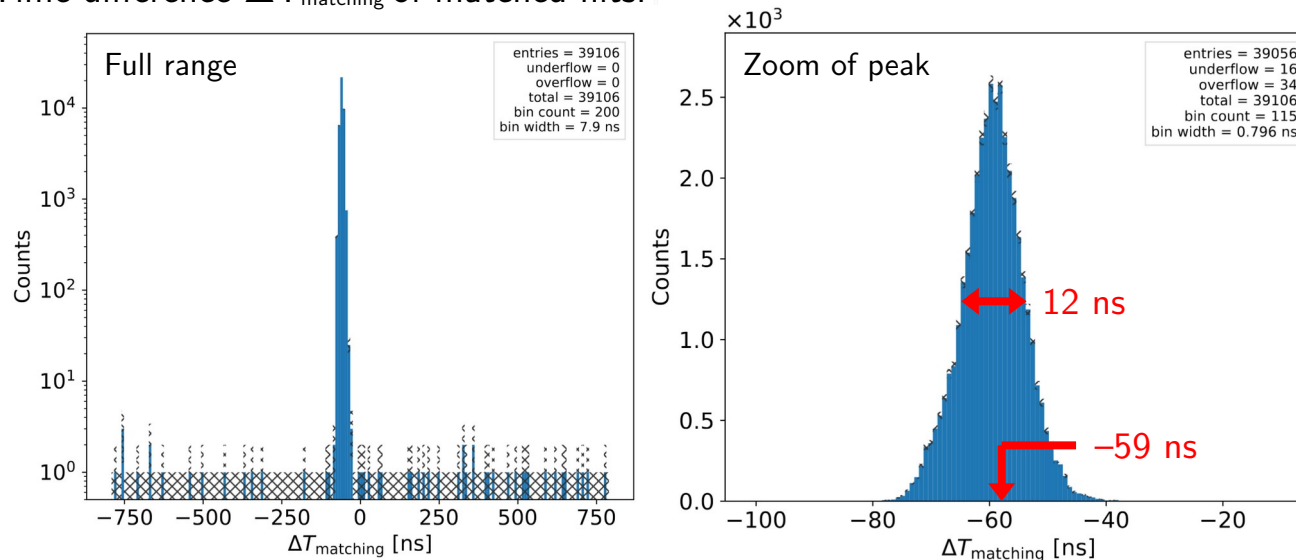
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- Scintillator layer rate: 8 Hz

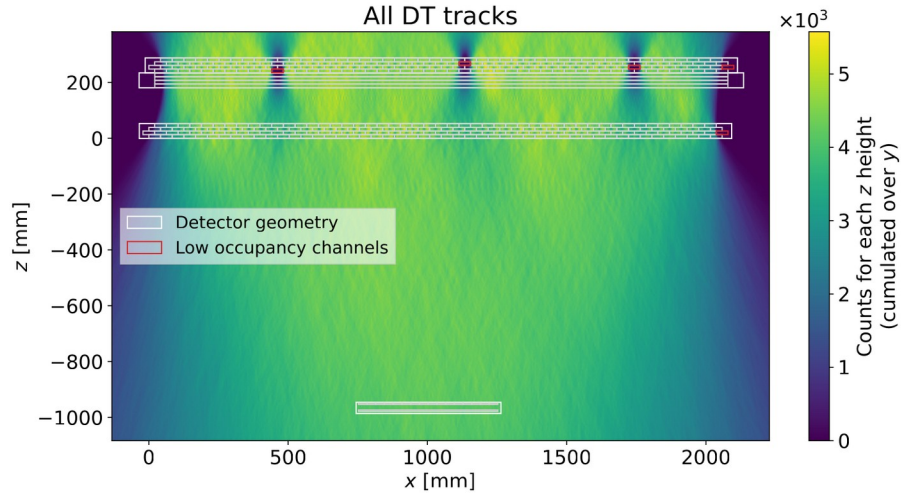
Time difference $\Delta T_{\text{matching}}$ of matched hits:



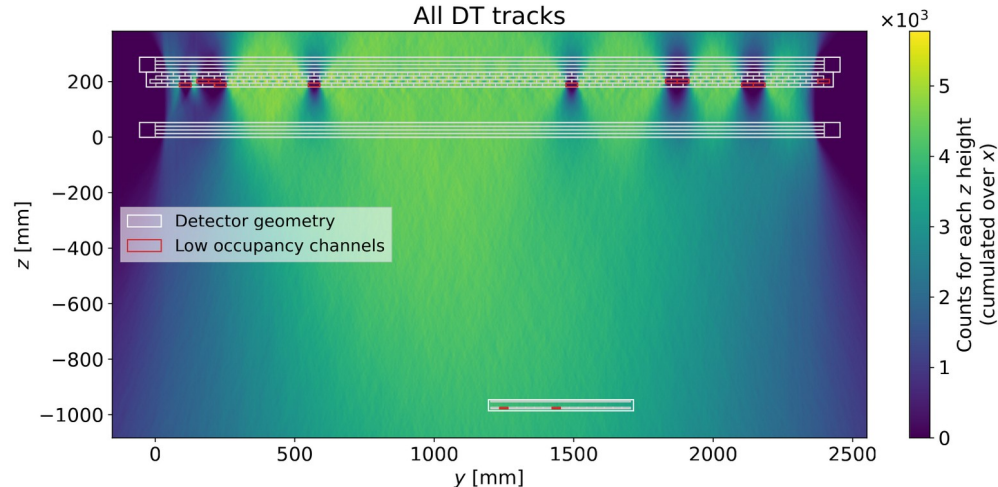
► **Successful detector synchronization !**

- **Peak FWHM: 12 ns**
- Includes all timing uncertainties of readout & reconstruction
- Observe constant **latency offset**
 - DT tracks are -59 ns later than scintillator hits
 - Possibly due to readout propagation difference

Independent cross-check of DT track reconstruction

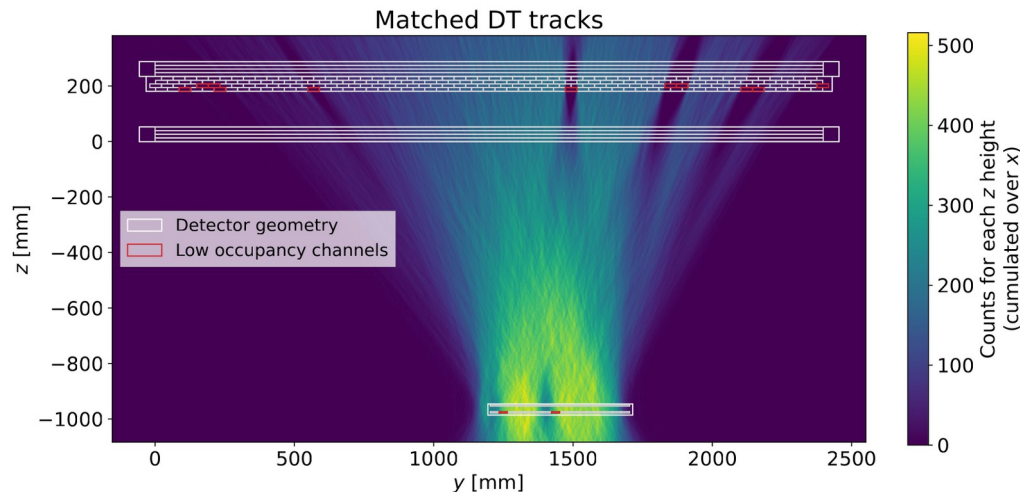
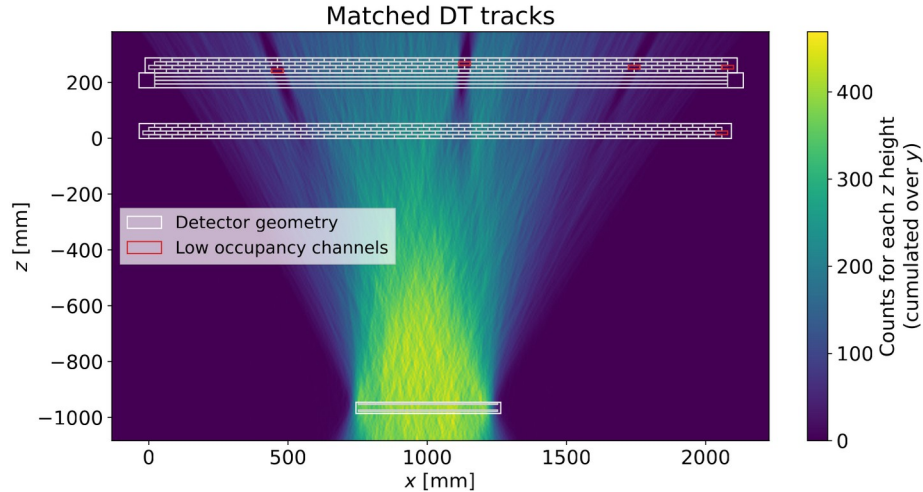


- Cross-check of DT track reconstruction
 - Analyze position of matched tracks w.r.t. scintillator
 - Unbiased, as tracks were only matched in time



Density of matched reconstructed tracks, in lateral view.

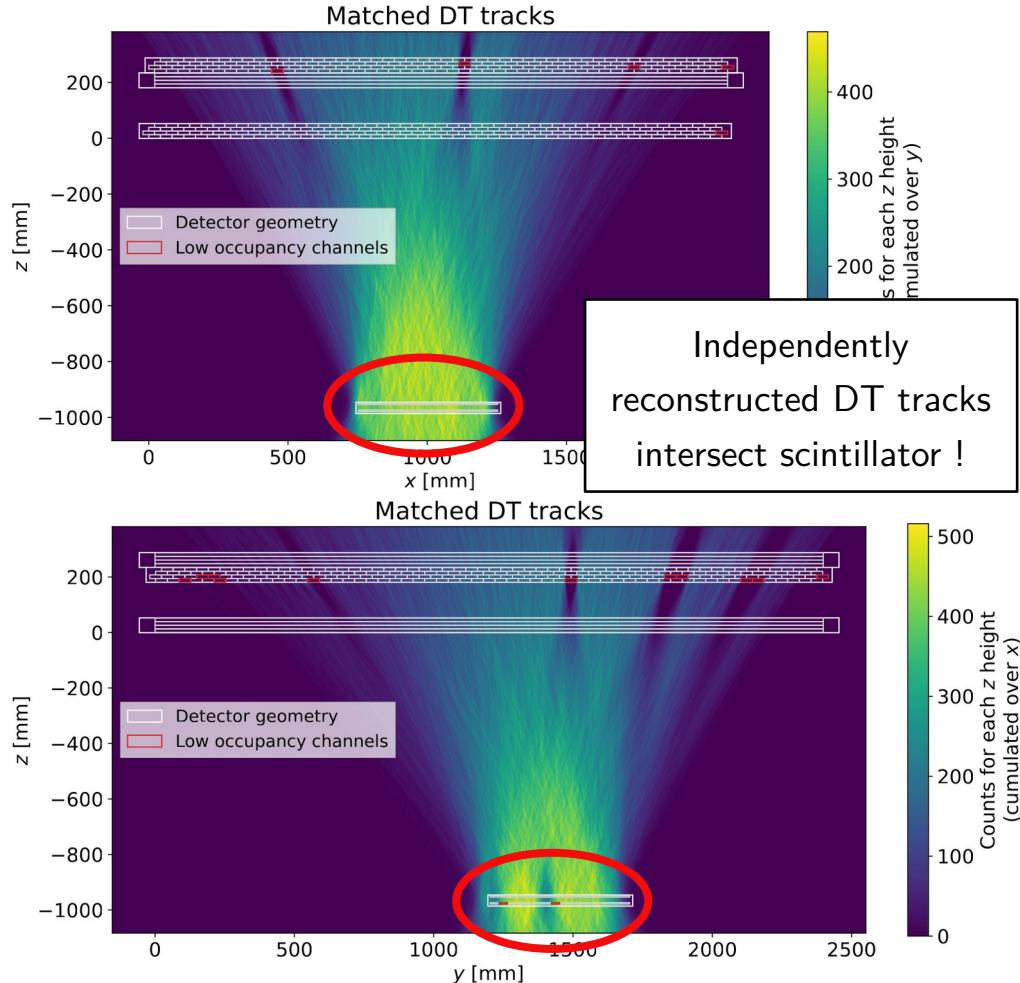
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Density of matched reconstructed tracks, in lateral view.

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- ▶ **Matched tracks are intersecting scintillator as expected !**
 - ▶ DT track reconstruction seems to work properly
 - ▶ Visible effect of DT cells & scintillator strips
- ▶ Need for proper **detector alignment**

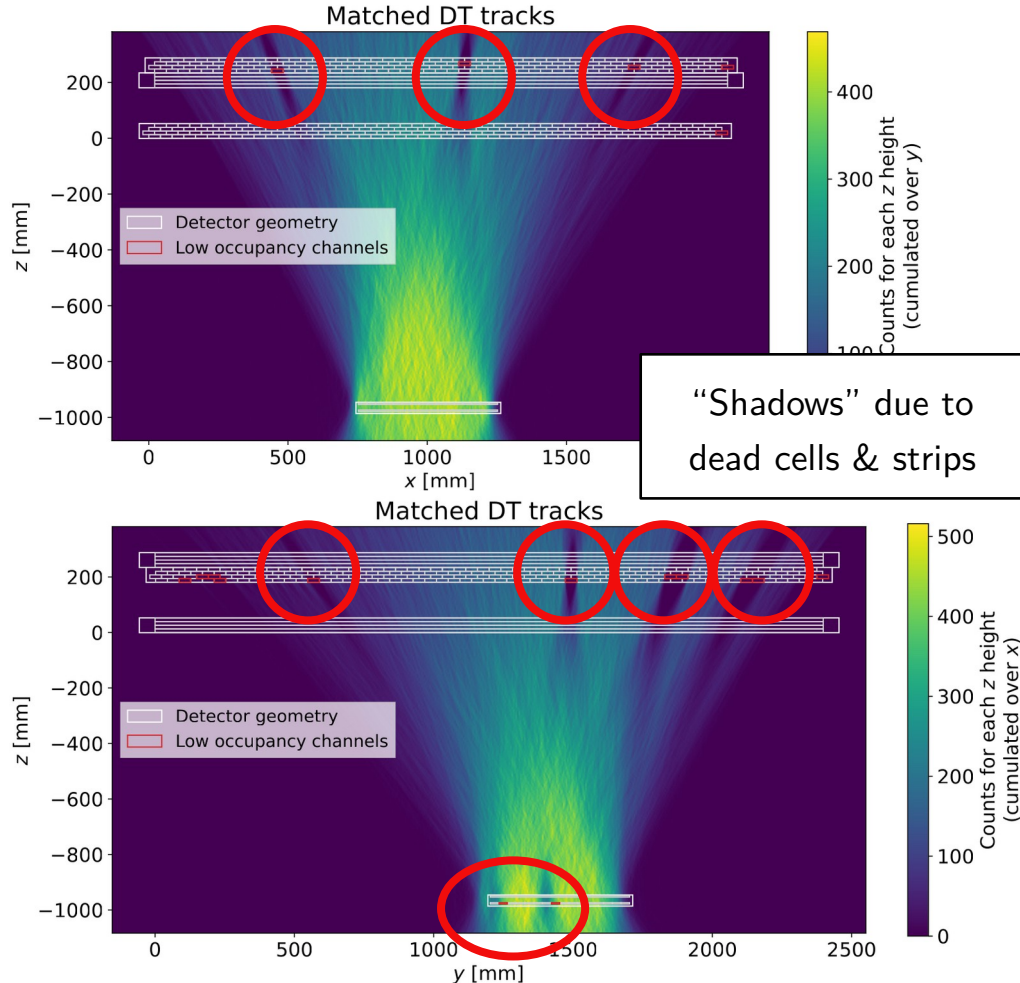
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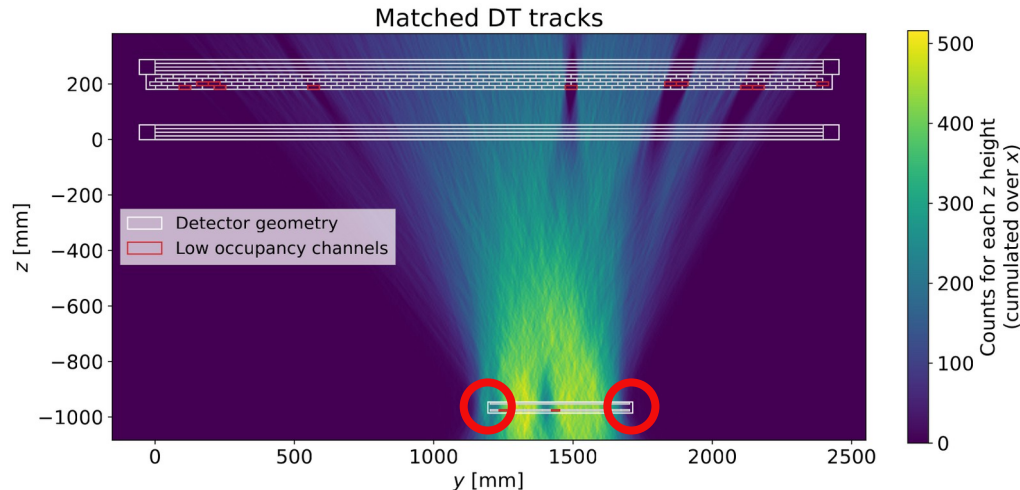
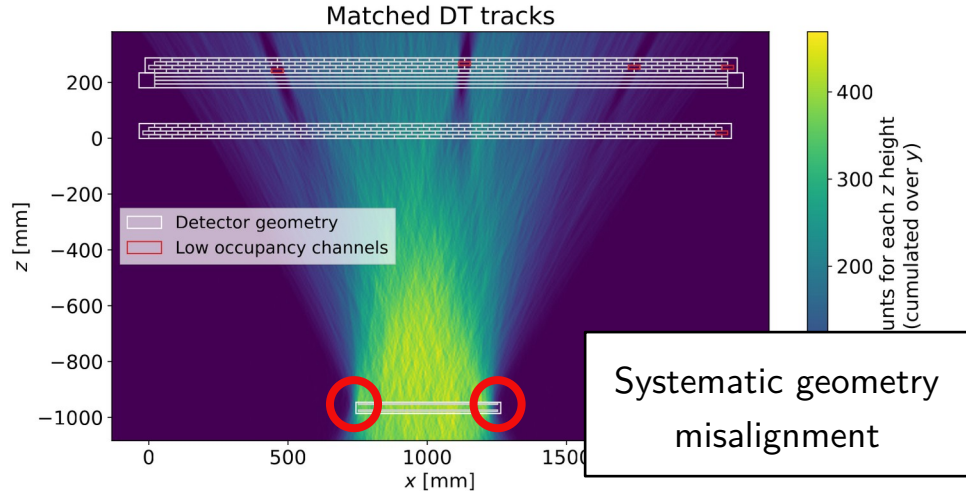
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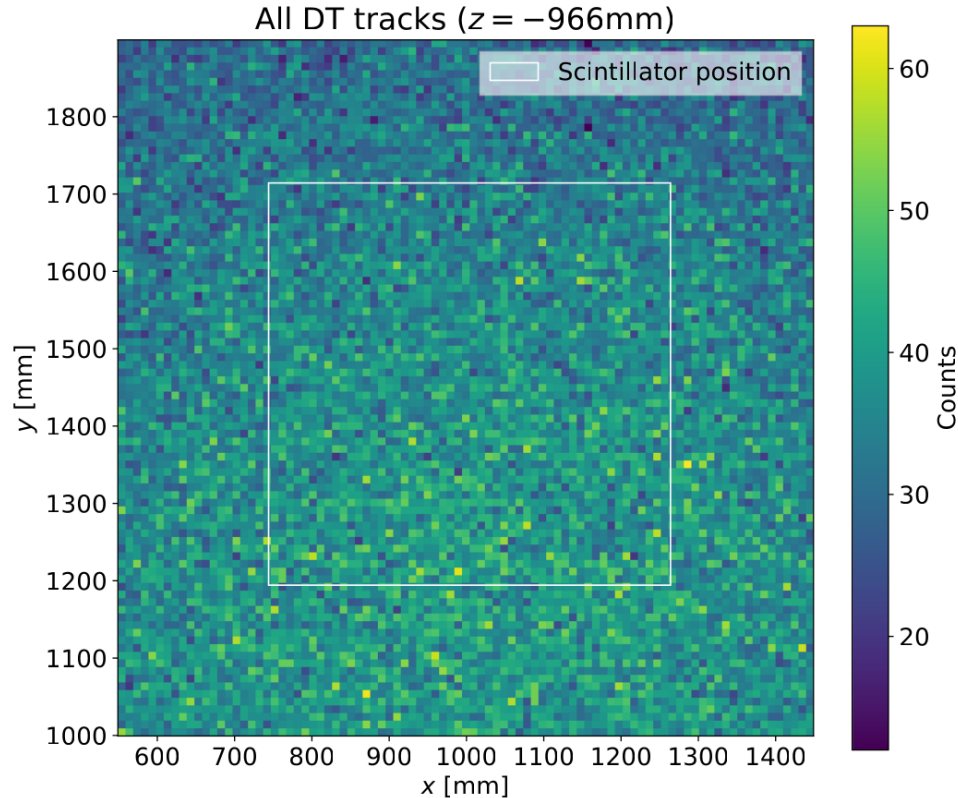
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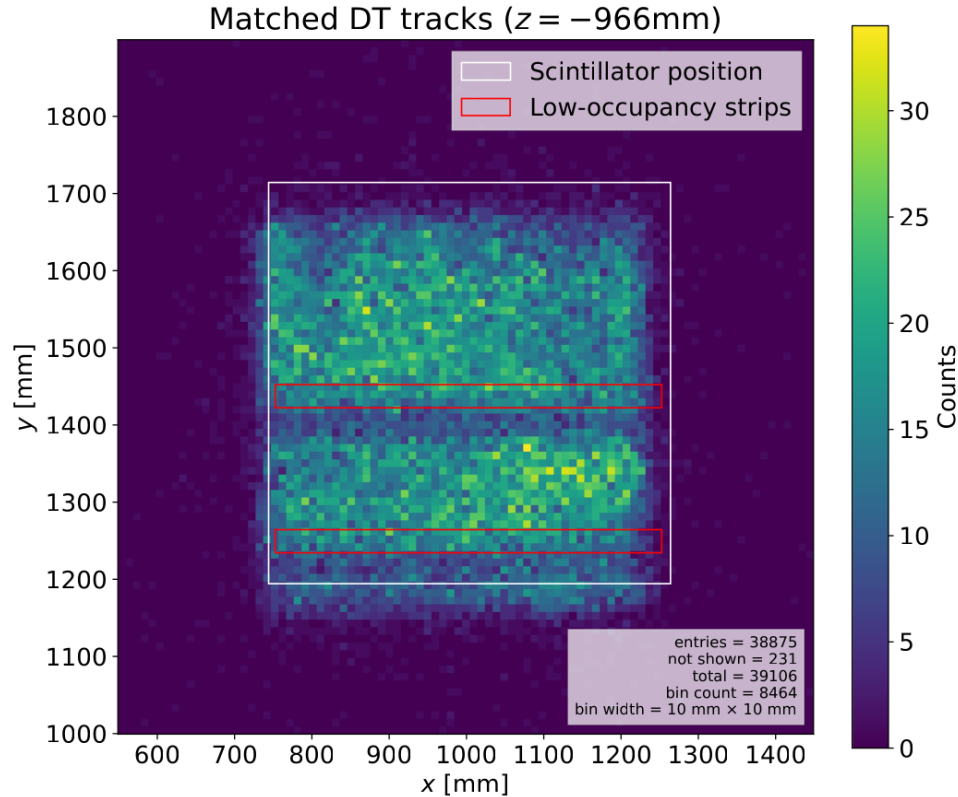
2D histogram of matched tracks in scintillator:



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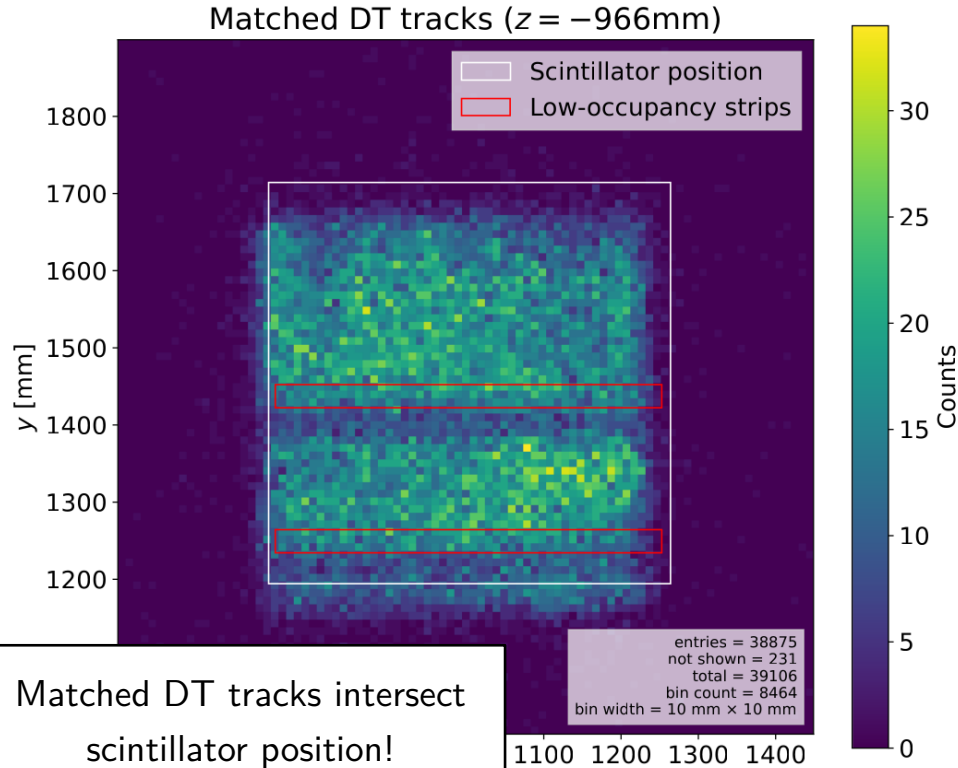
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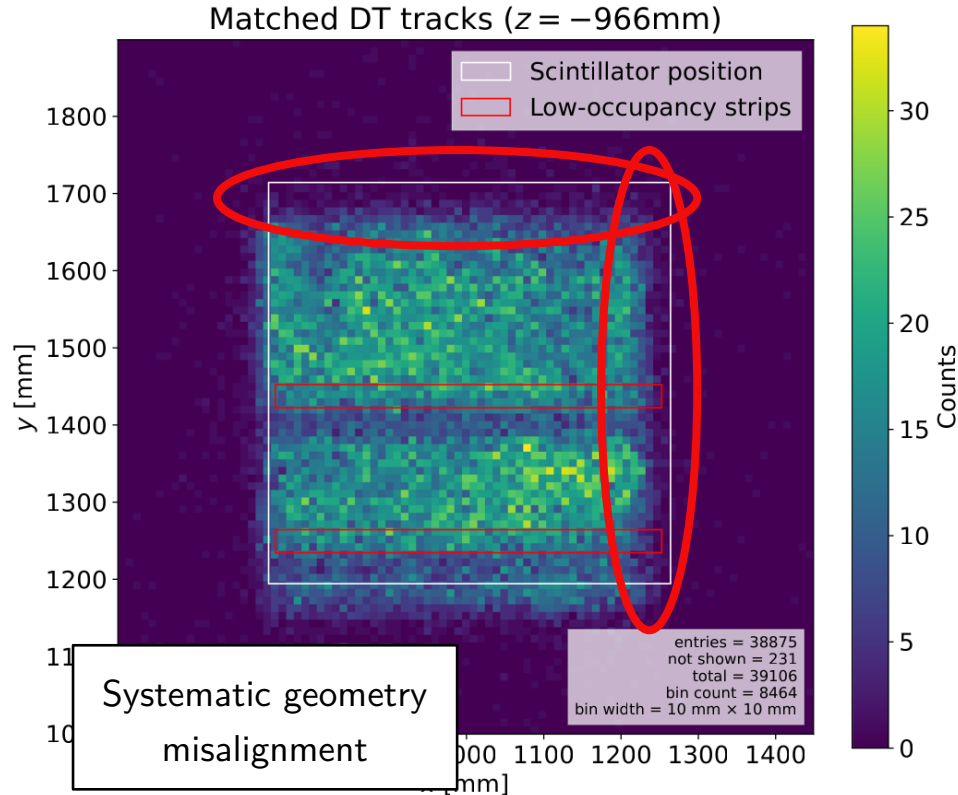
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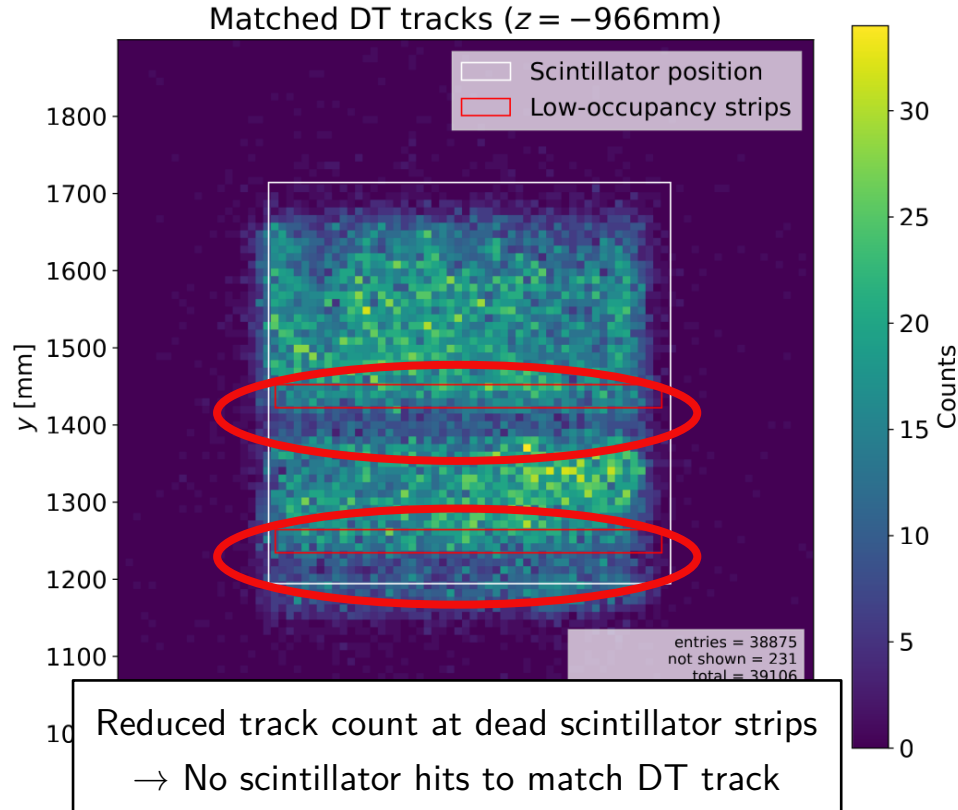
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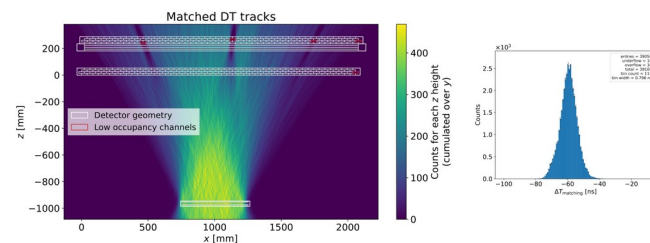
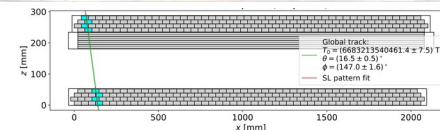


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Summary & outlook

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- DT chamber at cosmic test stand:
 - Successful installation & commissioning of CMS DT Phase-2 electronics
 - Implementation of triggerless track reconstruction algorithm
- Scintillator strip detector:
 - Developed new readout system & firmware (featuring IpGBT chip)
 - Observed systematic effects in SiPM readout and established mitigation with dead time
- Combined analysis:
 - Detector synchronization $O(10\text{ ns})$ was achieved, including all timing uncertainties
 - Positive independent cross-check of DT track reconstruction



Outlook:

- ▶ Further investigate systematics in scintillator readout system
- ▶ Optimize triggerless DT reconstruction algorithm & investigate possibility to use online DT trigger algorithm
- ▶ Provide detector alignment procedure
- ▶ Gained experience can be used to convert DT chamber to cosmic test stand for other detectors (beyond the scintillator)

Thank you for your attention!

Questions and comments are of course welcome.

Thank you!

I would like the following (incomplete) list of people who made this thesis possible:

- **Prof. Dr. Alexander Schmidt** for the opportunity and support to write my thesis at his institute
- **Apl. Prof. Dr. Oliver Pooth** for kindly agreeing on being the second corrector
- **Dr. Markus Merschmeyer** for his support and valuable feedback
- **Dr. Dmitry Eliseev** for supervising this thesis and his tireless support
- **Carsten Presser** for contributing several important aspects
- **Prof. Dr. Thomas Hebbeker** for his interest in my work and the helpful suggestions
- (...)

Backup

- 1) Theory
- 2) DT chamber description & data analysis
- 3) Scintillator description & data analysis
- 4) Combined data analysis
- 5) Simplified Monte Carlo simulation of cosmics

Backup (Pt. 1)

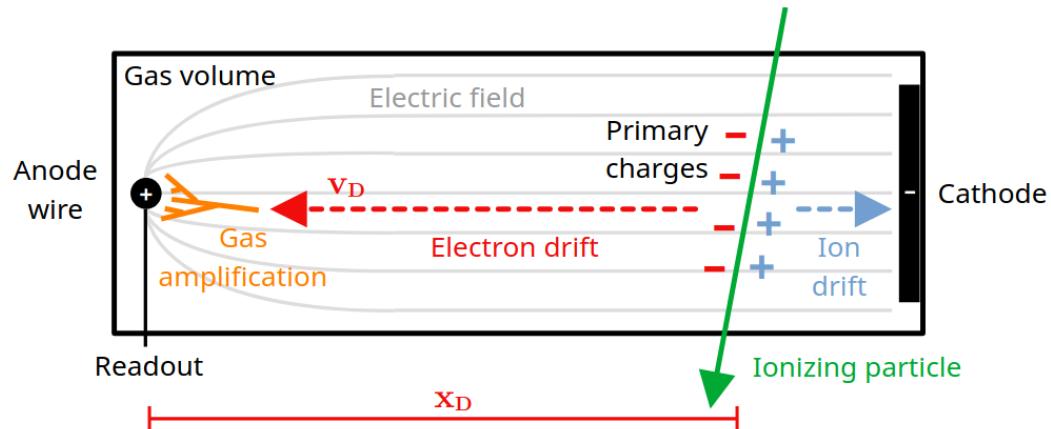
- Theoretical background
- Principles of particle detection

Atmospheric muons

- Cosmic rays interact with Earth's atmosphere
- Muons originate from decay of light hadrons
- Vertical muon flux: $\Phi = 70 \text{ m}^{-2} \text{ s}^{-1} \text{ sr}^{-1}$
- Zenith angle dependence: $\Phi(\theta) \sim \cos^2 \theta$

Drift chamber

- Ionization of gas in detector volume from passing muon
- Separation of charges in electric field
- Charge drift with constant velocity
- Gas amplification in proximity of wire
- Initial hit position can be reconstructed from drift time measurement
 - Resolution far beyond the drift cell granularity

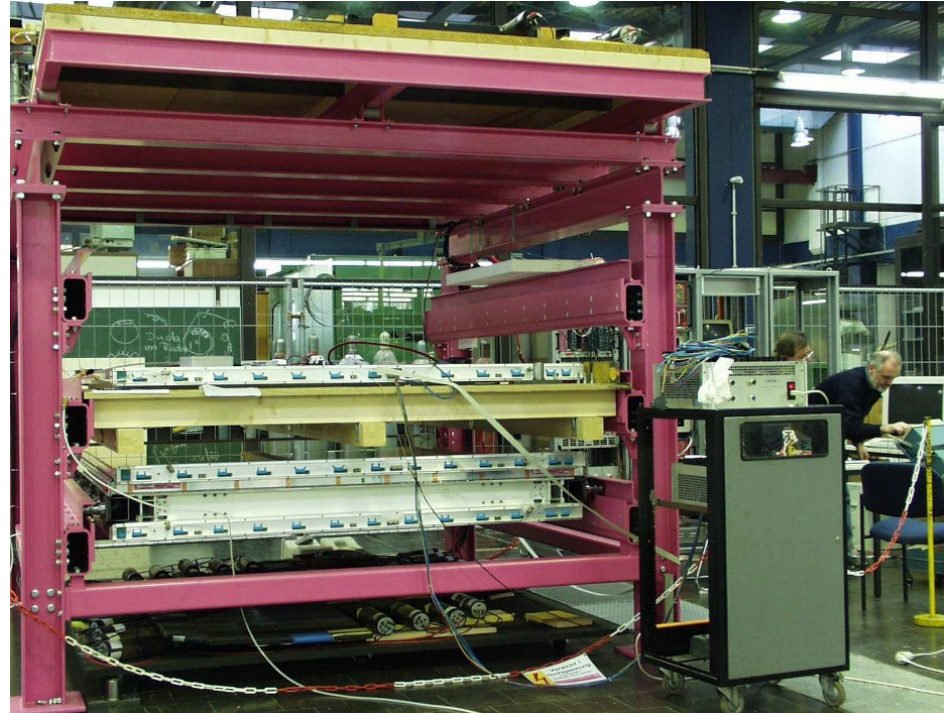


Backup (Pt. 2)

- CMS DT system
- Triggerless DT data analysis

Cosmic test stand in early 2000s

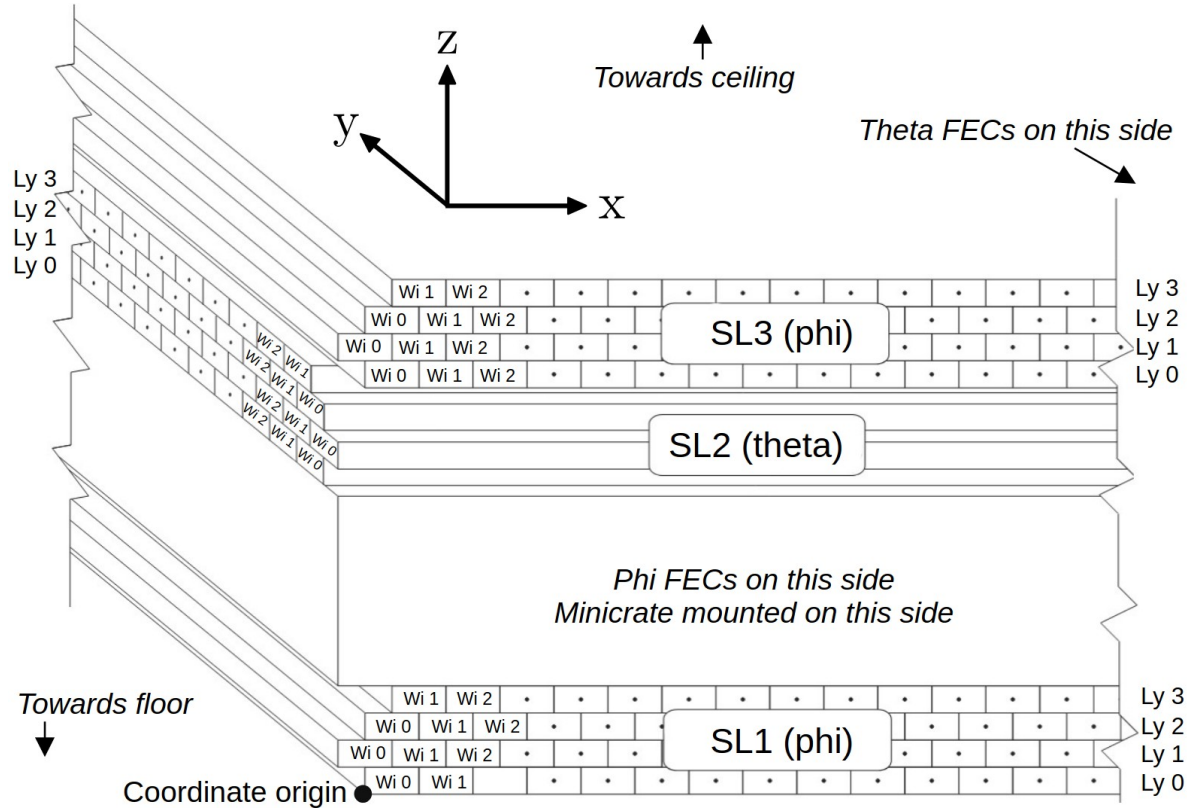
Used for quality control of DT chamber production at our institute:



- ▶ Trigger scintillators above & below chamber
- ▶ Custom TDCs for drift time measurement

DT global coordinate frame

Selected global coordinate frame in this work:



DT readout electronics

DT chamber frontend board:

- Signal discrimination: 35 mV threshold
- Output of digital hit pulse

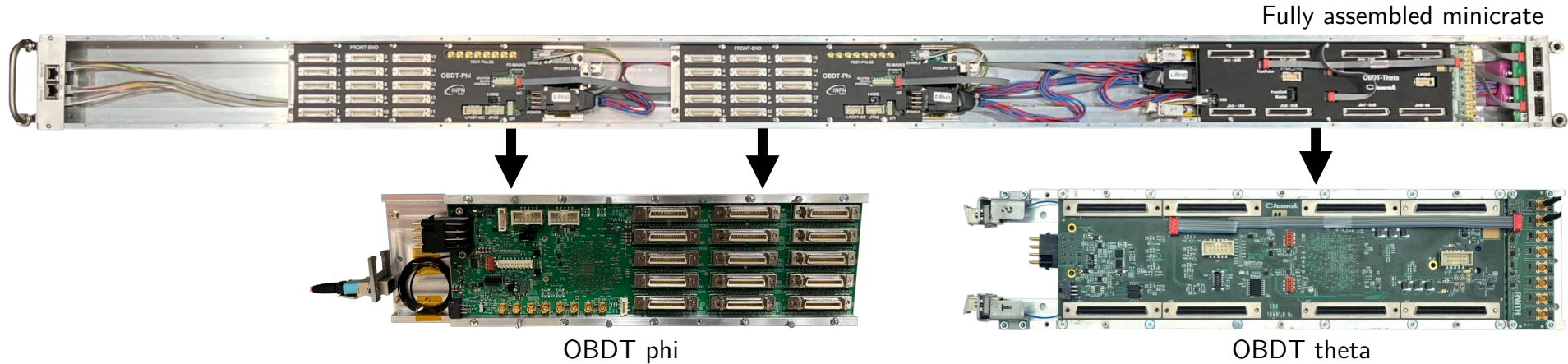


Frontend board

DT readout electronics

Minicrate (on-chamber readout electronics):

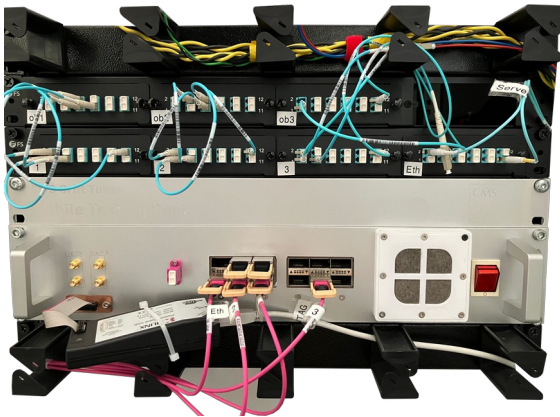
- On-board Electronics for Drift Tubes (OBDDT) boards in two flavors (phi & theta)
 - Hit digitization with 0.78 ns granularity
 - Data transmission over 10.24 Gb/s optical links



DT readout electronics

Backend system:

- Prototype backend hardware: “Slow control & timing box” (SC&T box)
 - No online trigger algorithms, only stores raw DT hits
- Actual CMS DT Phase-2 backend system not available yet

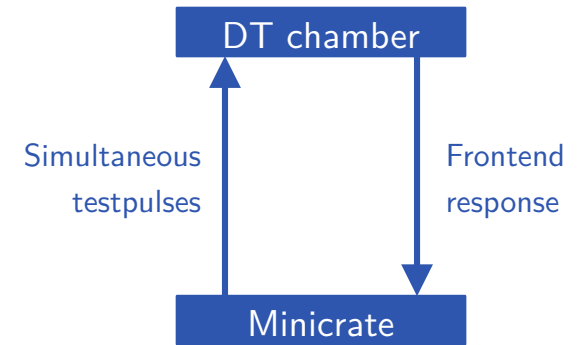
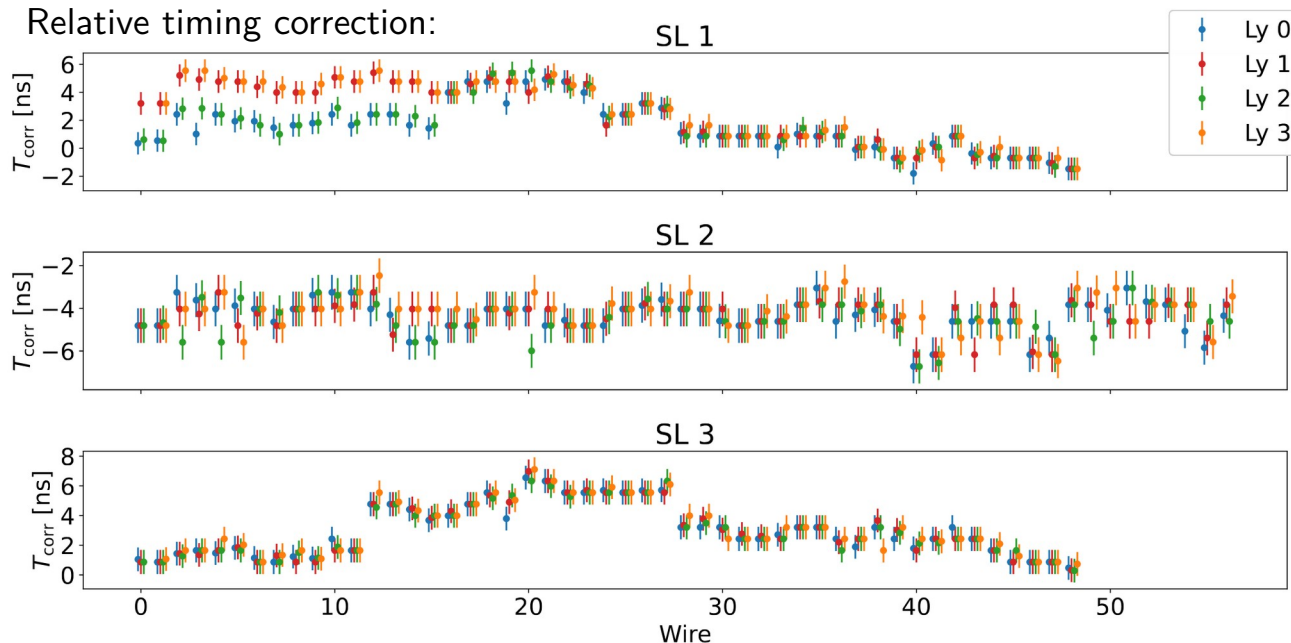


Optical fiber patch panel

SC&T box backend

DT timing calibration with testpulses

- DT reconstruction relies on cell hit timestamps with $O(\text{ns})$ precision
 - Relative timing calibration is necessary
- Minicrate generates simultaneous testpulses
 - Well-defined timing of sent testpulse
 - Record response timestamp and estimate relative correction T_{corr}

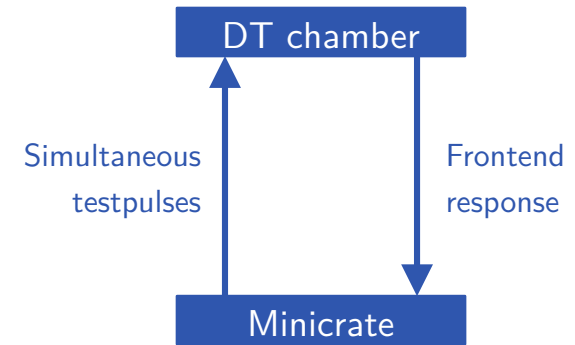
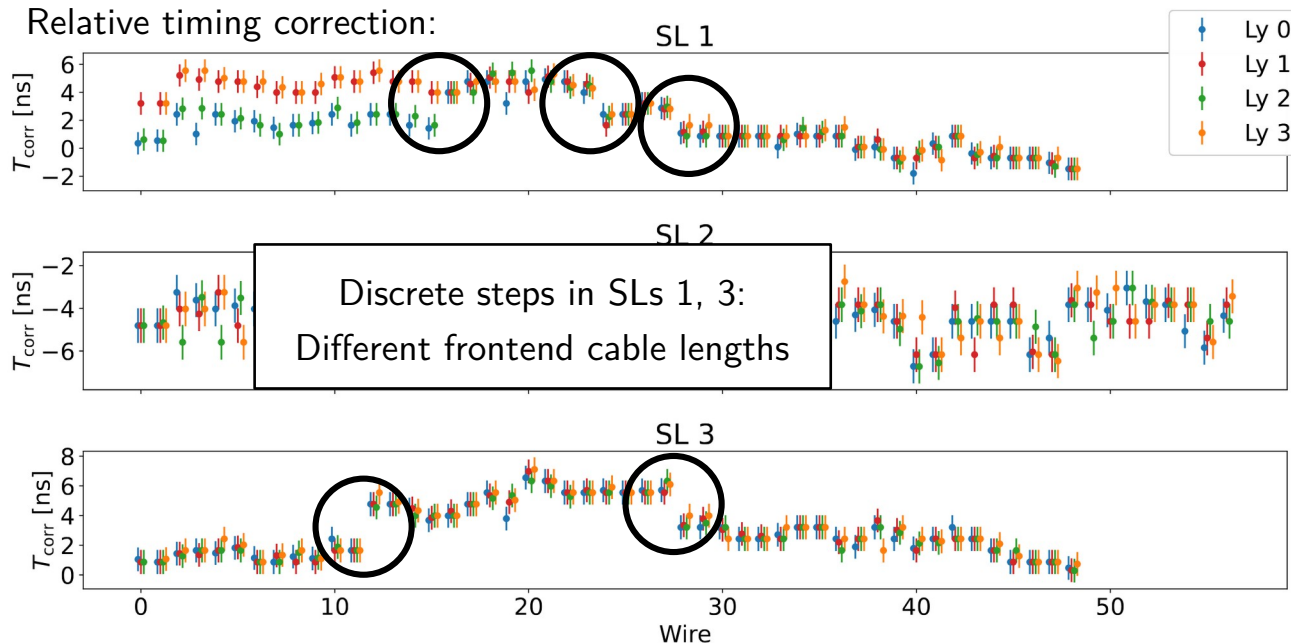


$$T_{\text{corr}} = T_{\text{response}} - \langle T_{\text{response}} \rangle$$

- ▶ Obtain DT timing corrections in the expected range (< 10 ns)
- ▶ Are applied in offline data analysis

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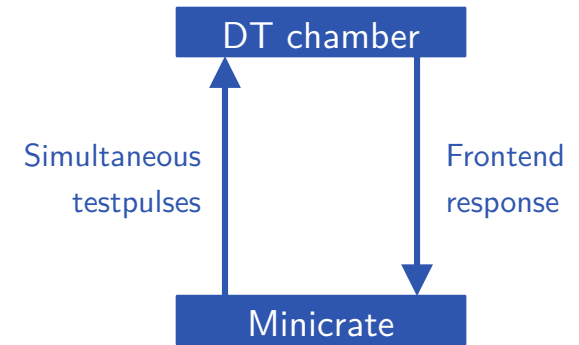
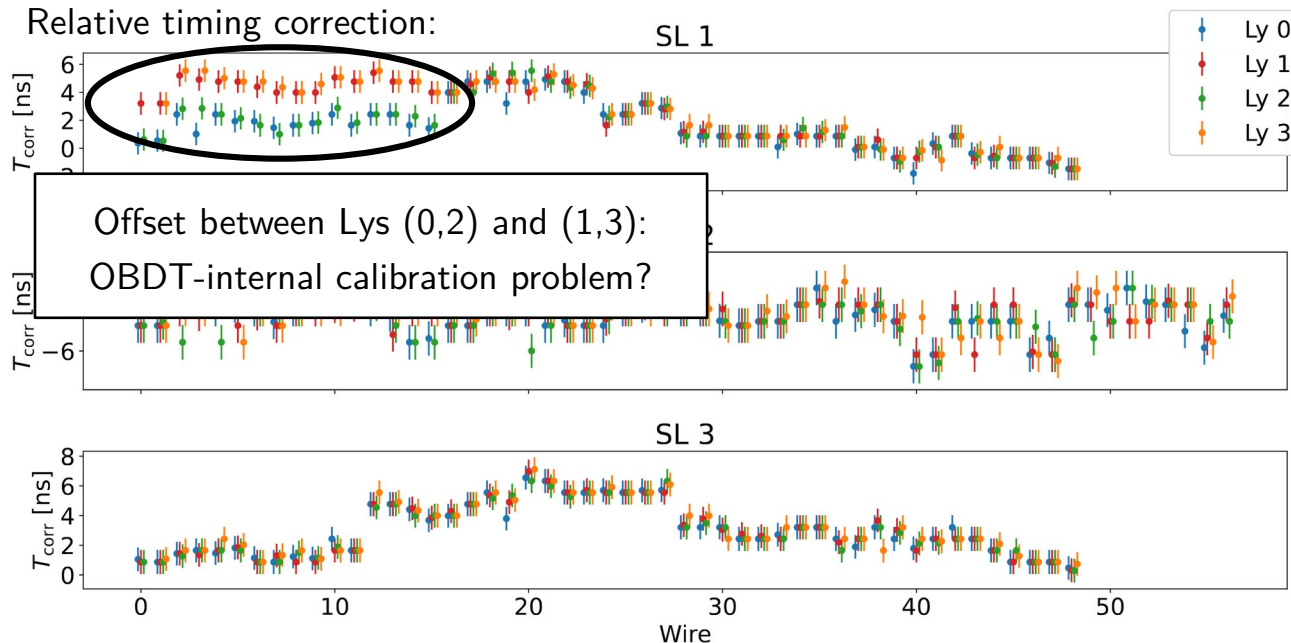


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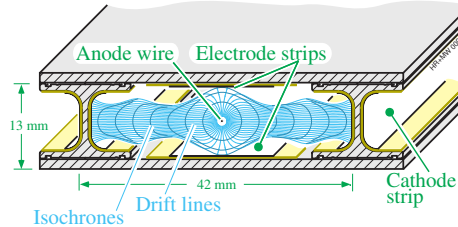
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DT secondary hits

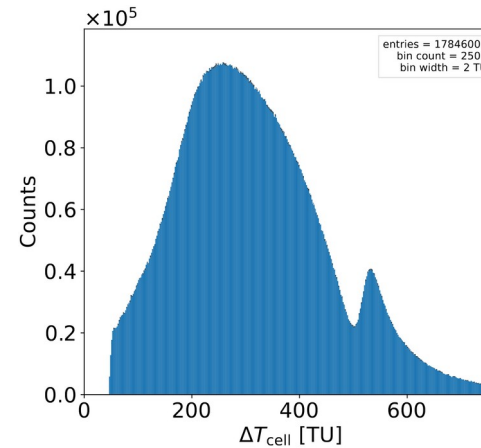
- Time difference between consecutive hits of same cell reveals presence of secondary hits
 - Remove Poisson background events
 - Observe double peak structure in < 500 ns range
- Similar structure was observed in CMS publications

Physical cause:

- Photoeffect leads to secondary ionization
 - Photons from avalanche near anode wire create photoelectrons at electrodes
 - Peak 1: Field shaping electrodes
 - Peak 2: Cathode electrodes ($t_{\text{drift}} \approx t_{\text{drift,max}}$)

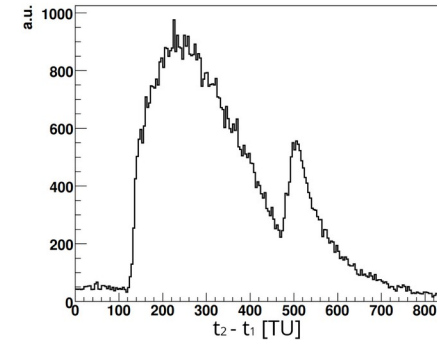


Time difference between hits:



1 TU (timestamp unit) = 0.78 ns

Comparison with
CMS-NOTE-2008-003:



- Apply individual cell dead time of 468 ns to suppress secondary hits in the recorded dataset

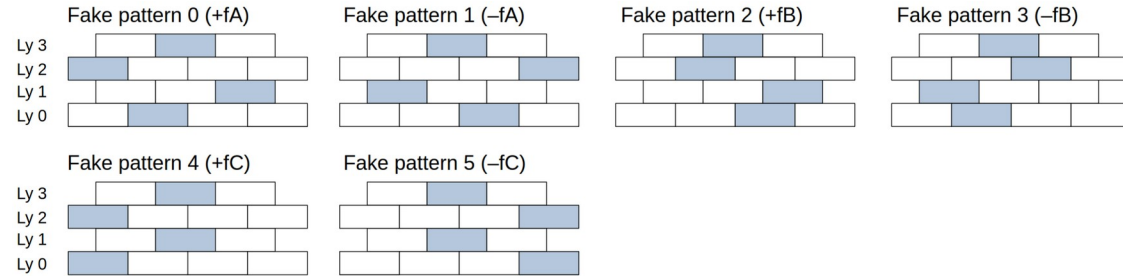
Estimation of drift velocity: $\mathbf{v_{drift} = d_{half-cell} / (t_{peak\ 2} - t_{initial}) \approx (50.5 \pm 0.2) \mu m/ns}$

- Doubts about precision: Need detailed simulation to understand all involved effects
- Therefore continue analysis with CMS reference value: $v_{\text{drift}} = 54.5 \mu\text{m/ns}$

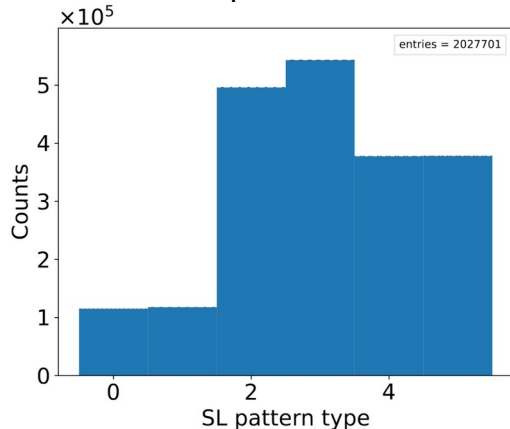
DT superlayer fake patterns

- Analyze influence of noise hits with fake patterns

Superlayer fake pattern geometry:



Pattern occupancies:

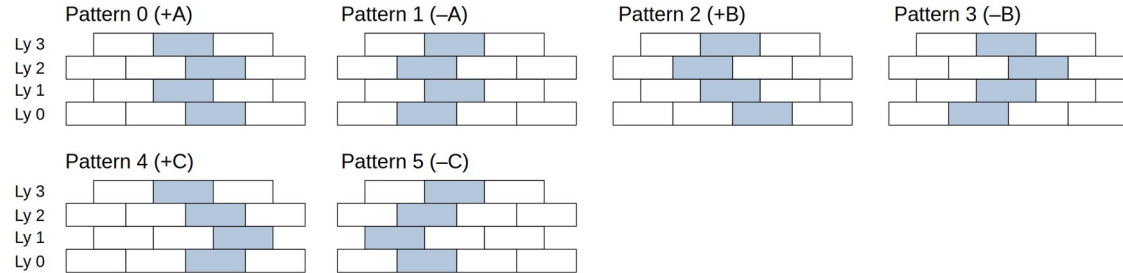


- Observe cumulated pattern rate of 35 Hz / superlayer

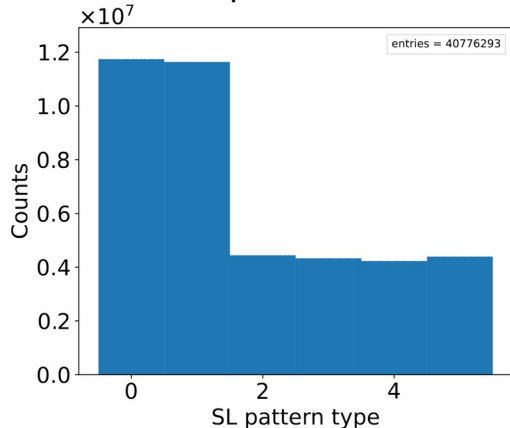
DT superlayer patterns

- Perform triggerless track reconstruction:
 - Treat all cell hits as part of potential track
- Superlayer pattern:
 - Require track-like shape of hits
 - Hit time difference limited by max. drift time:
$$\Delta t_{\text{max}} \leq t_{\text{drift,max}}$$
 - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



- Cumulated pattern rate: **700 Hz / superlayer**

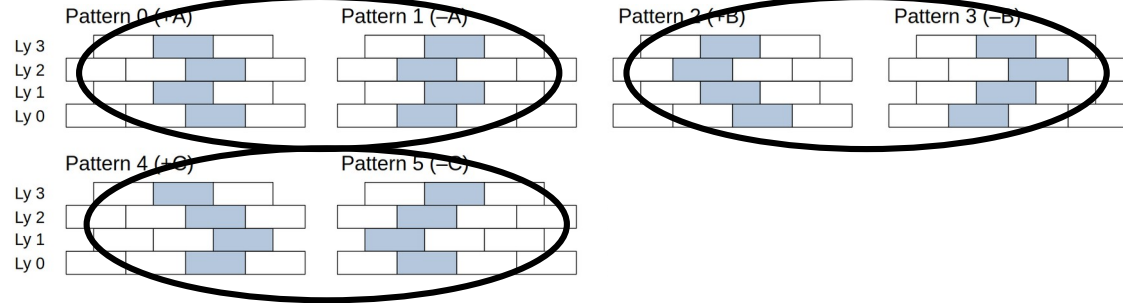
Atmospheric muon simulation: 450 Hz / superlayer

- Presence of noise-induced patterns in data

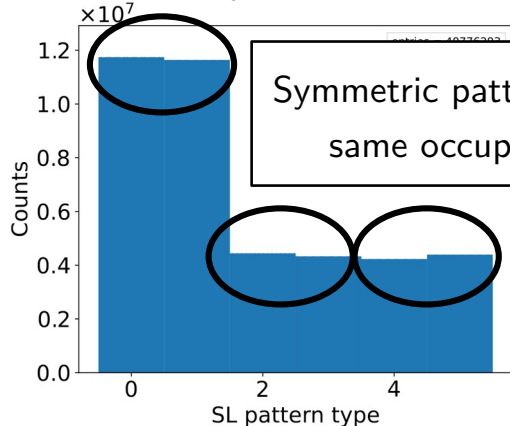
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Superlayer pattern geometry:



Pattern occupancies:



Symmetric patterns have same occupancies

Simulated pattern rate: **700 Hz / superlayer**

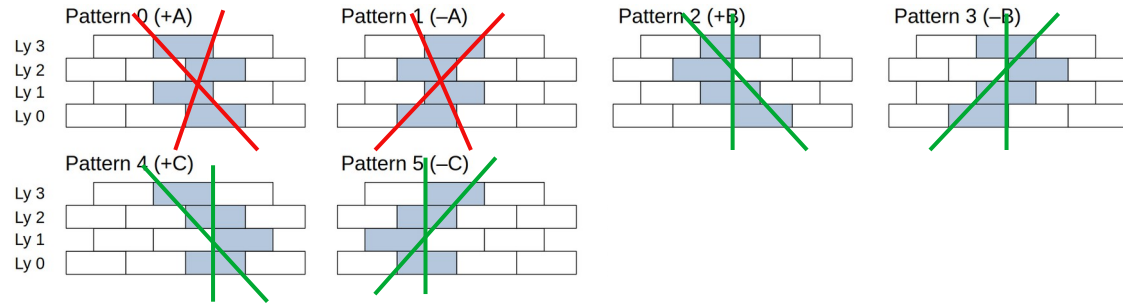
Atmospheric muon simulation: 450 Hz / superlayer

Presence of noise-induced patterns in data

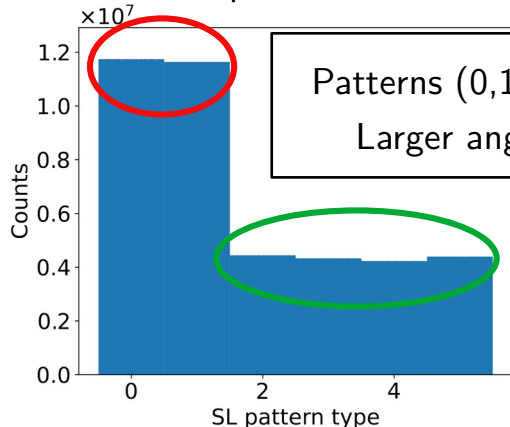
DT superlayer patterns

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 - Treat all cell hits as part of potential track
- Superlayer pattern:
 - Require track-like shape of hits
 - Hit time difference limited by max. drift time:
$$\Delta t_{\max} \leq t_{\text{drift},\max}$$
 - Only accept patterns with 4 cell hits

Superlayer pattern geometry:



Pattern occupancies:



Patterns (0,1) have higher rate:
Larger angular acceptance

pattern rate: **700 Hz / superlayer**

muon simulation: 450 Hz / superlayer

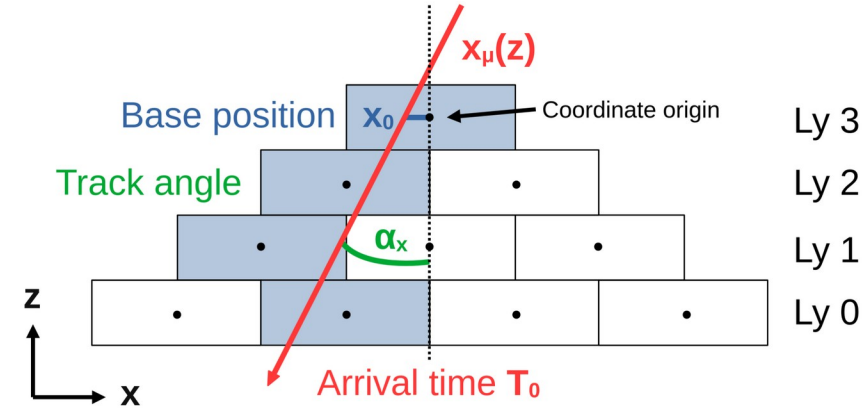
► Presence of noise-induced patterns in data

DT superlayer pattern fit

- Assumptions for reconstruction:
 - Constant drift velocity over full length: $\mathbf{v}_{\text{drift}} = \text{const.}$
 - Horizontal drift at height of wire: $\mathbf{z}_{\text{hit},ly} = \mathbf{z}_{\text{wire},ly}$
 - Straight muon tracks: $\mathbf{x}_{\text{hit},ly} = \mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha$
- Relation between cell hit position $\mathbf{x}_{\text{hit},ly}$ and timestamp $\mathbf{T}_{ly}$:
 - Arrival time $\mathbf{T}_0$ determines drift time: $\mathbf{t}_{\text{drift},ly} = \mathbf{T}_{ly} - \mathbf{T}_0$
 - Drift distance: $\mathbf{x}_{\text{drift},ly} = \mathbf{v}_{\text{drift}} \cdot \mathbf{t}_{\text{drift},ly}$
 - Left-right ambiguity of cell hit described by laterality $\mathbf{L}_{ly}$:

$$\mathbf{x}_{\text{hit},ly} = \mathbf{L}_{ly} \cdot \mathbf{x}_{\text{drift},ly} + \mathbf{x}_{\text{wire},ly}$$
- Fit function of 4 timestamps ($\mathbf{T}_{ly}$) to 3 track parameters ($\mathbf{T}_0$, $\mathbf{x}_0$, $\tan \alpha$):

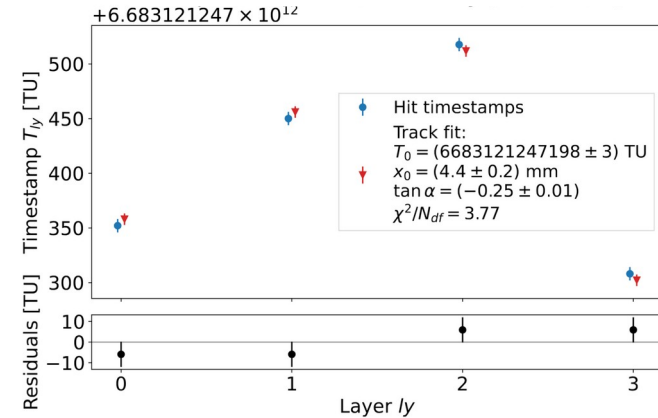
$$\mathbf{T}_{ly} = \mathbf{T}_0 + \mathbf{L}_{ly} / \mathbf{v}_{\text{drift}} \cdot (\mathbf{x}_0 + \mathbf{z}_{\text{wire},ly} \cdot \tan \alpha - \mathbf{x}_{\text{wire},ly})$$



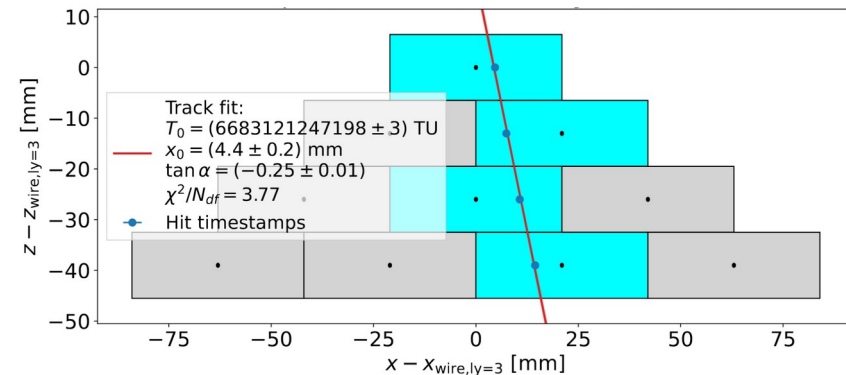
DT superlayer pattern fit

- Perform timestamp fits for all individual patterns
 - Consider all laterality combinations and keep only best fit
- Apply cuts to fit results to suppress noise hits:
 - Goodness-of-fit variable: $\chi^2/N_{df} < 10$
 - Unphysical drift times: $2 \text{ ns} \leq t_{\text{drift}} \leq t_{\text{drift,max}} - 2 \text{ ns} = 383 \text{ ns}$
- ▶ Fitted pattern rate: **400 Hz / superlayer**
 - ▶ Atmospheric muon simulation: 450 Hz / superlayer
 - ▶ Relatively good agreement

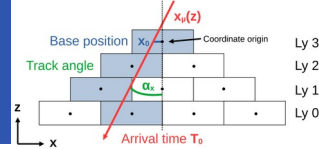
Example timestamp fit result:



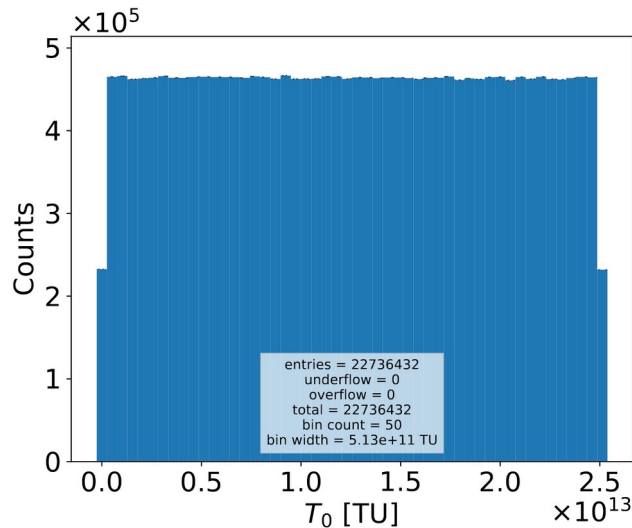
Reconstructed cell hit positions:



DT superlayer pattern fit results

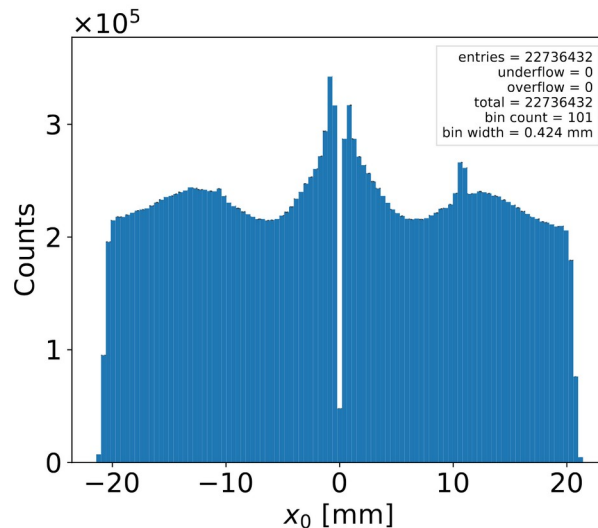


Arrival time T_0 :



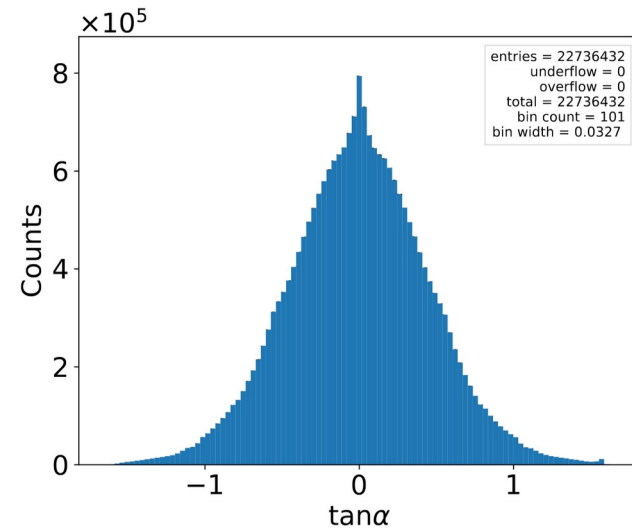
- ▶ Uniform distribution as expected for atmospheric muons

Reference position in cell x_0 :



- ▶ Roughly uniform distribution in cell
 - ▶ Increased number of tracks close to wire: Field nonlinearity
 - ▶ Small feature at $x_0=10.5$ mm: Noise hits?

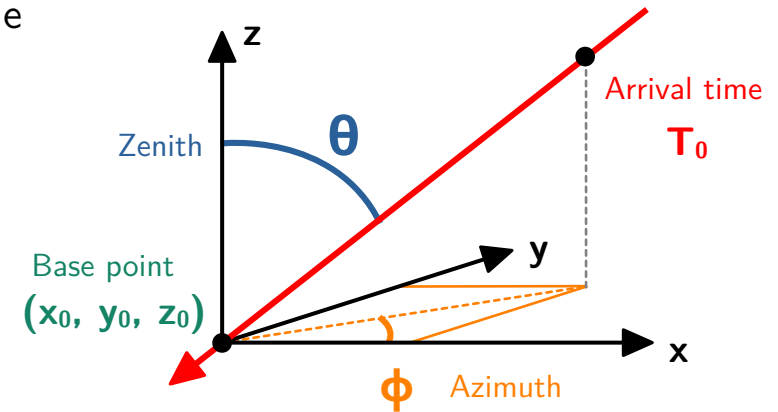
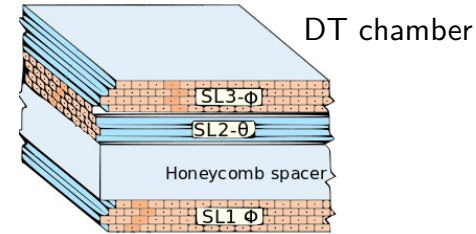
Track inclination $\tan \alpha$:



- ▶ Vertical tracks are preferred
- ▶ Symmetric shape
 - ▶ Small feature at $\tan \alpha = 0$: Noise hits?

DT global track building

- Temporal matching of pattern fits in different superlayers
 - Select patterns close in time: $\Delta T_{0,SL} \leq 20 \text{ ns}$
 - Remove pattern ambiguities
 - Check consistency of SLs 1, 3 pattern fits
- Combination of superlayer projections into global coordinate frame
 - Global track parameters:
 - Arrival time T_0
 - Base point (x_0, y_0, z_0)
 - Azimuth angle Φ , zenith angle Θ
- ▶ Reconstructed global muon rate: **85 Hz**
- ▶ Global track parameters (mostly) behave as expected

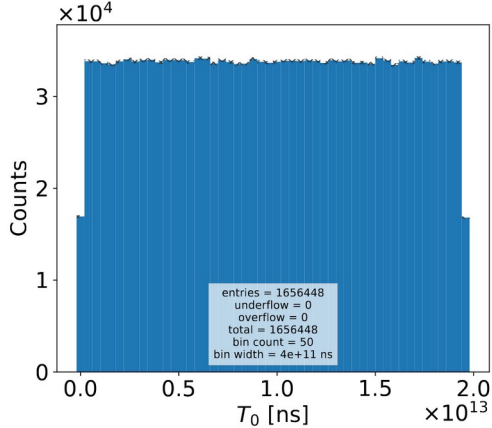


Atmospheric muon simulation: 196 Hz

- ▶ Lower rate of reconstructed muons: Possibly because of assumptions in reconstruction

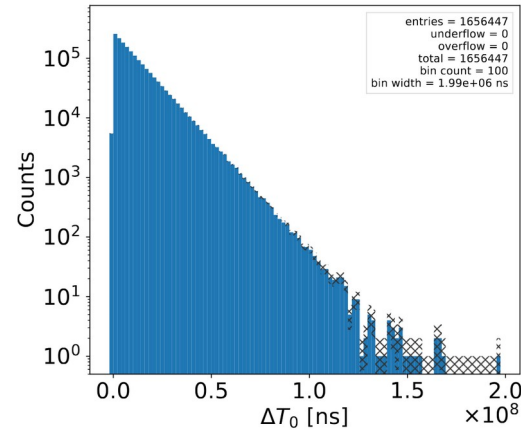
DT global track results

Arrival time T_0 :



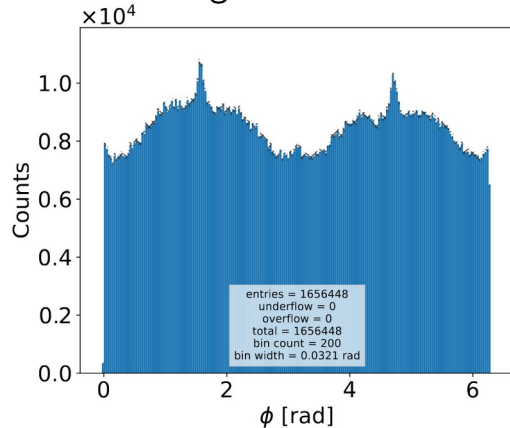
- ▶ Uniform distribution as expected for atmospheric muons

Time difference ΔT_0 of consecutive tracks:



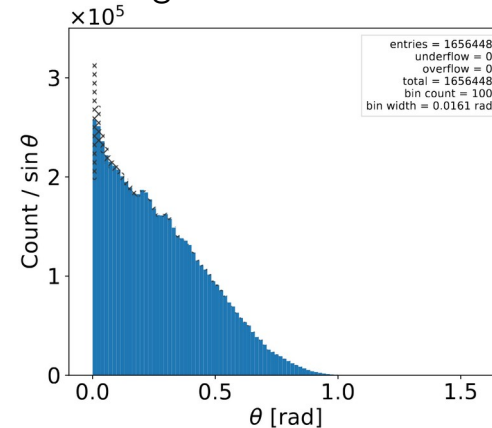
- ▶ Poisson-random arrival times of atmospheric muons leads to observed exponential distribution for time difference

Azimuth angle Φ :



- ▶ Symmetry between $[0, \pi]$ and $[\pi, 2\pi]$ as expected
- ▶ Spike at $\Phi = \pi/2, 3\pi/2$: Noise hits?
- ▶ Non-uniform due to cuboid chamber shape

Zenith angle Θ :



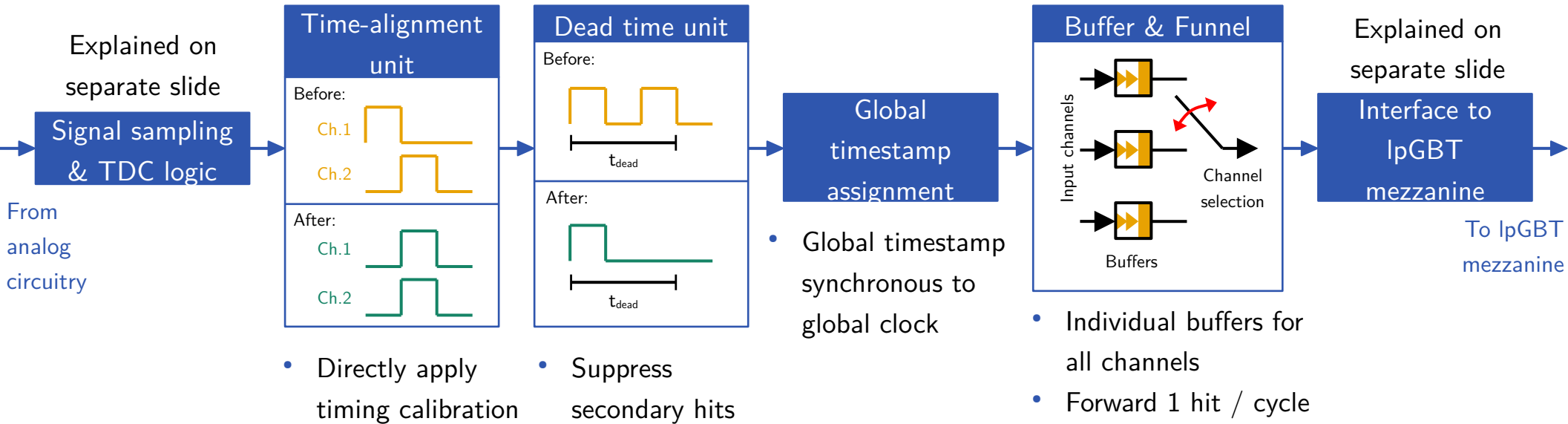
- ▶ Vertical tracks preferred
 - ▶ Matches qualitative expectation
- ▶ Steep decrease towards higher angles

Backup (Pt. 3)

- SiPM-based scintillation detector
- Scintillator data analysis

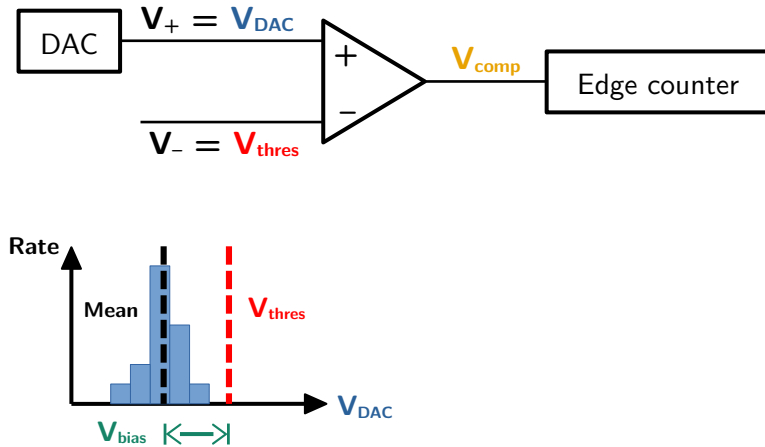
Scintillator readout electronics – FPGA firmware

- Completely self-developed readout firmware running on FPGA:

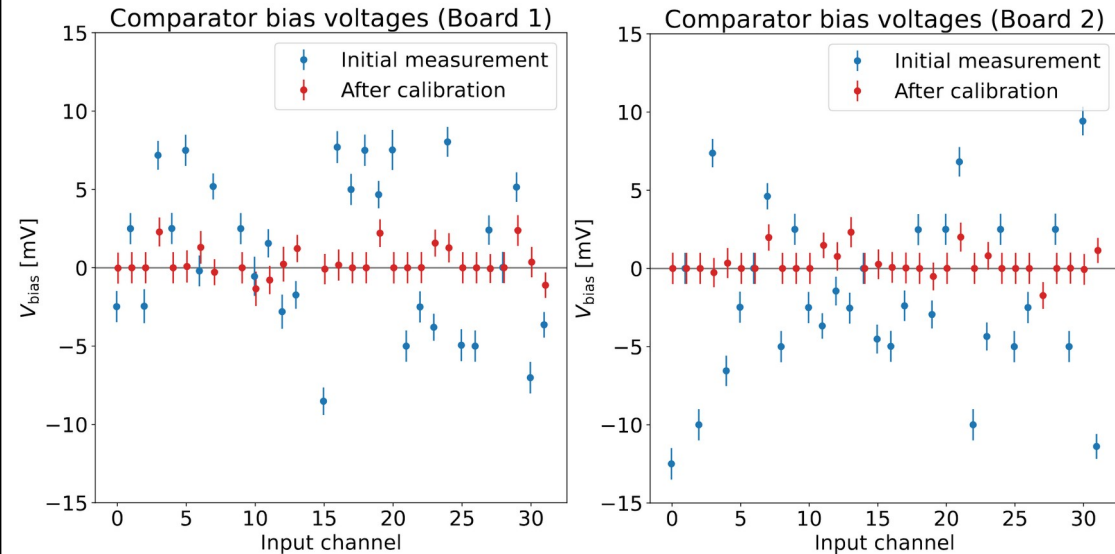


Input discriminator threshold calibration

- FPGA-internal comparators are non-ideal
 - Switching point shifted by bias voltage V_{bias}
- Bias measurement:
 - Scan comparator input voltage for fixed threshold
 - Determine comparator switching rate
 - Switching point lies at rate maximum

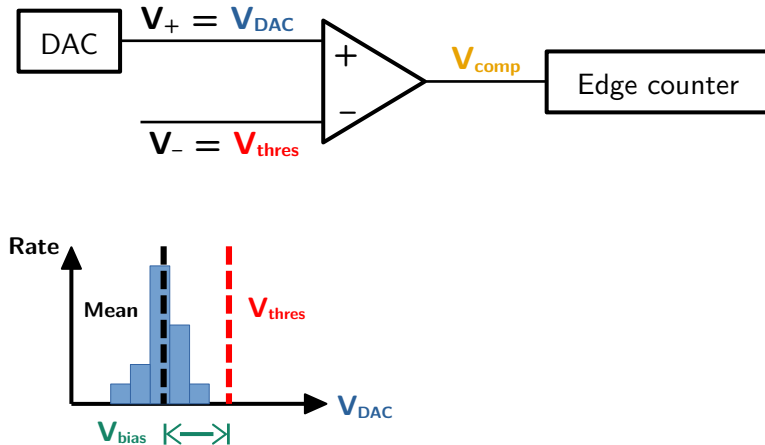


Measured bias before and after correction:

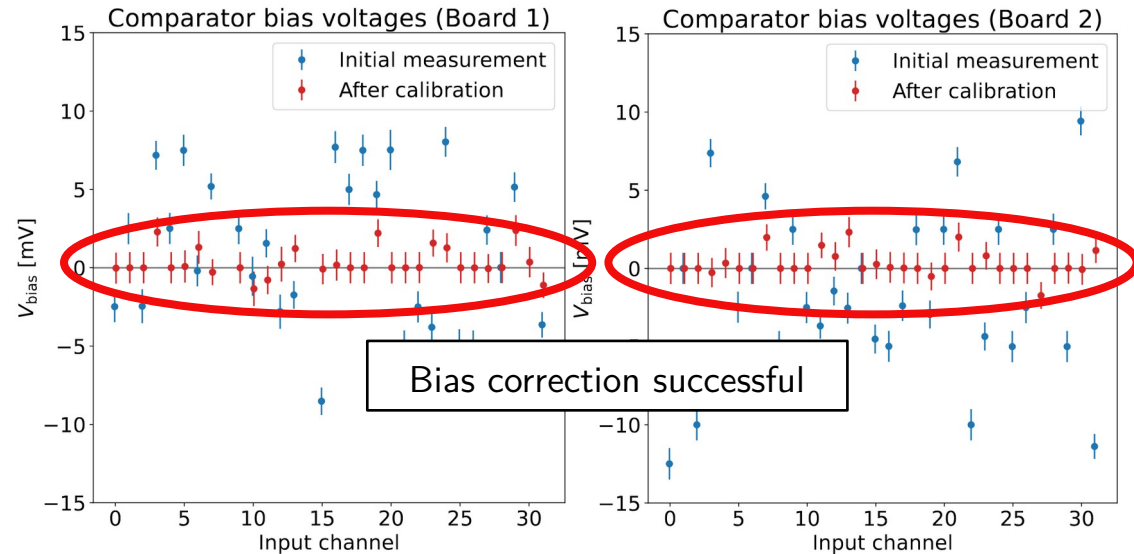


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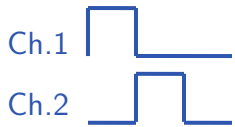
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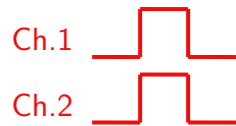
Input timing calibration with testpulses

- TDC logic implemented in FPGA has non-deterministic timing offset
 - Due to chip-internal calibration of deserializers on startup
 - Affects groups of inputs (I/O banks)
- Timing offset measurement:
 - Send simultaneous testpulses to all inputs
 - Measure response timestamps
- Time-alignment unit:
 - Programmable delay units in FPGA firmware
 - Calculate target delays from testpulse timing

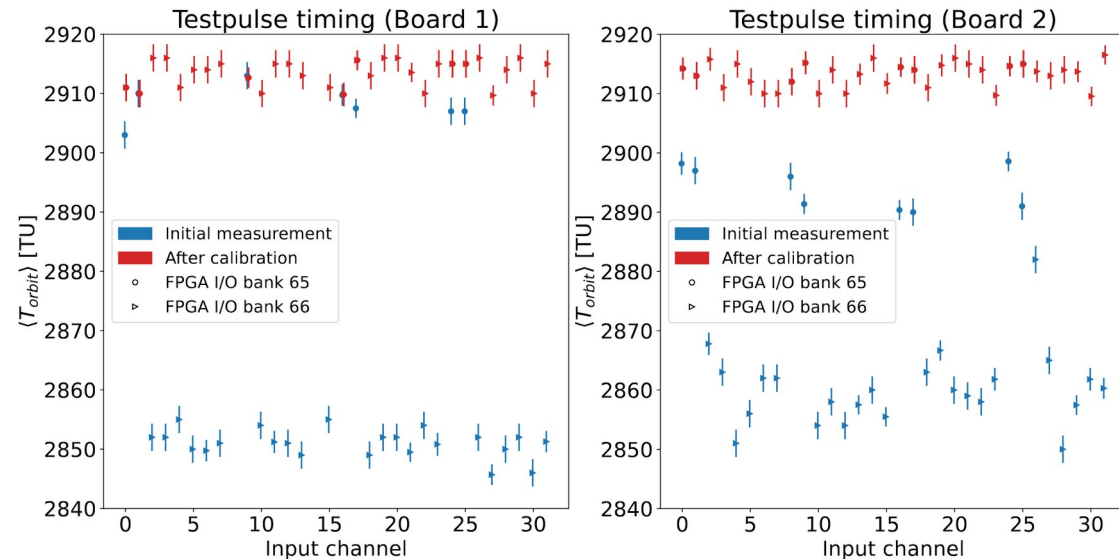
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

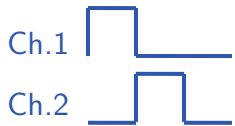


1 TU (timestamp unit) = 0.78 ns

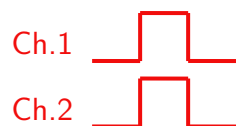
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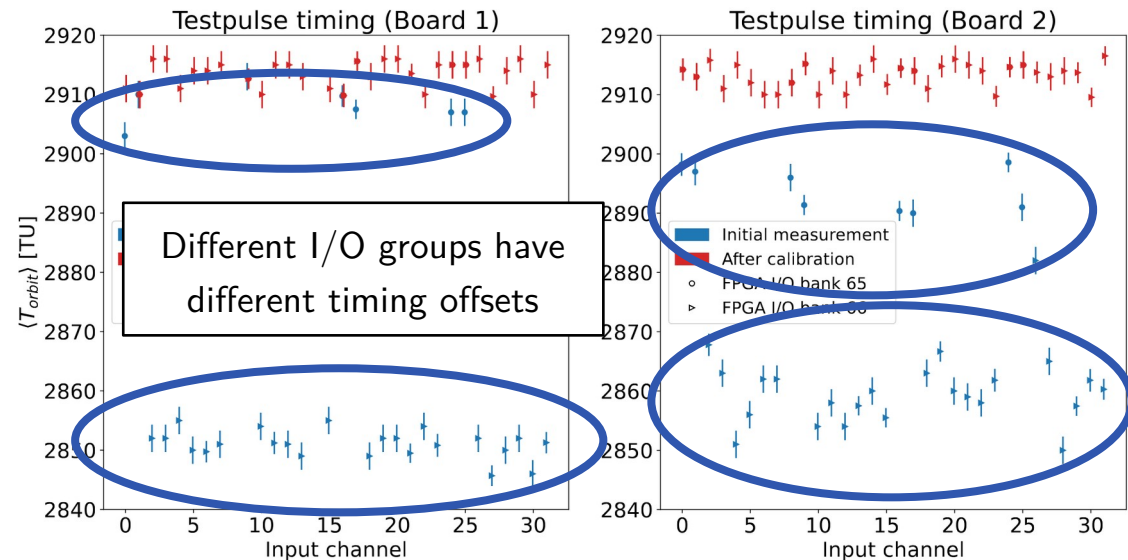
Before:



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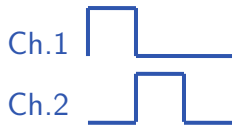


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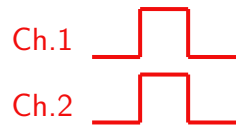
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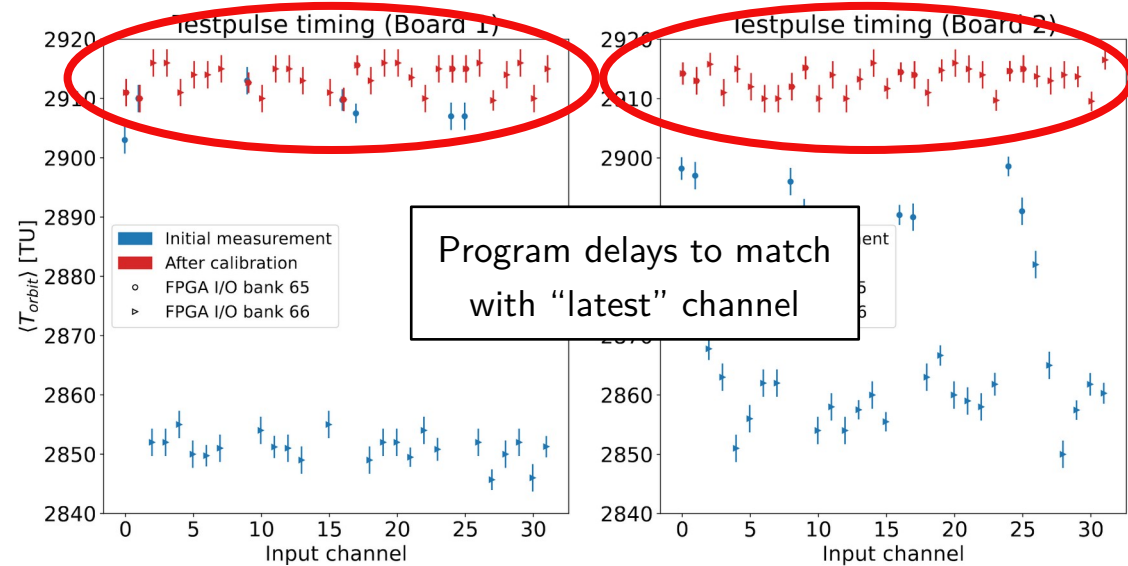
Before:



After:



- Measured testpulse response **before** and **after** programming delays:

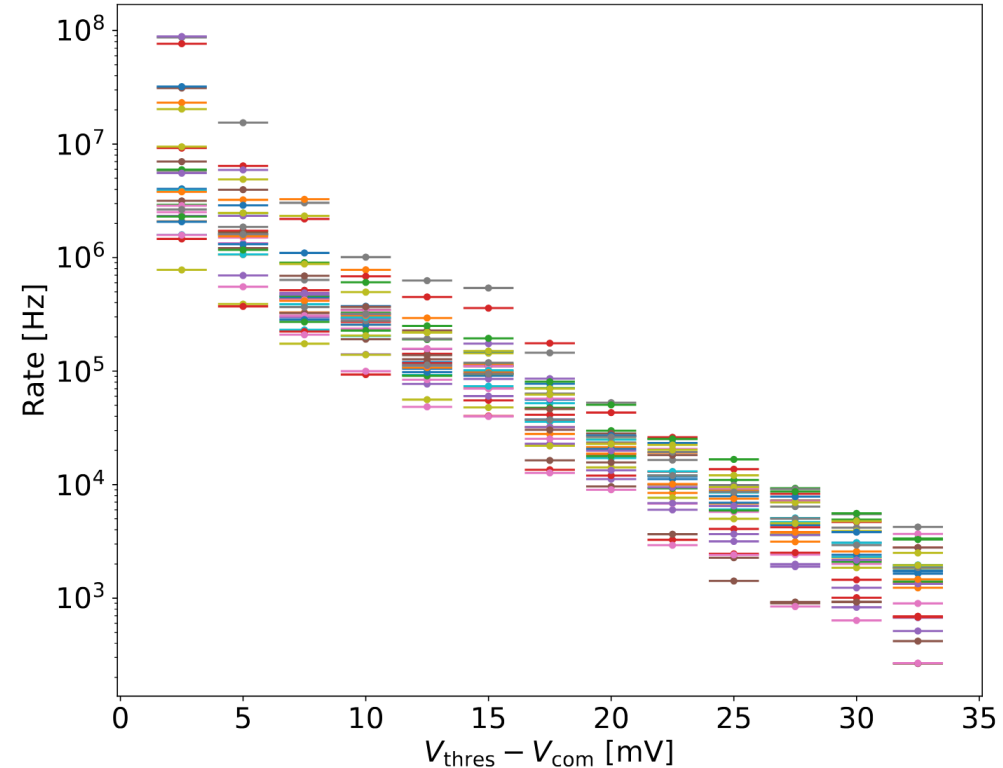


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SiPM noise scan

- Performed measurement of SiPM noise spectrum
 - Scan threshold voltage range [2.5, 32.5] mV (or [0, 6] p.e.)
 - Measure SiPM rate with counter directly on FPGA
- ▶ **Exponentially decreasing rate**
 - ▶ Expected from noise & cell crosstalk:
 - ▶ Randomly triggered cell due to thermal noise: Hit with amplitude $V_{\text{SiPM}} = 1$ p.e.
 - ▶ Crosstalk probability p with other cell allows higher amplitudes, however exponentially suppressed: $P(V_{\text{SiPM}} = N \text{ p.e.}) = p^N$
 - ▶ Discrete steps in rate due to integer cell count
 - ▶ Expected steps in noise rate not visible
 - ▶ Aliasing effects as 1 p.e. $\sim$ scan step size

Obtained rate spectrum for 29 SiPMs:

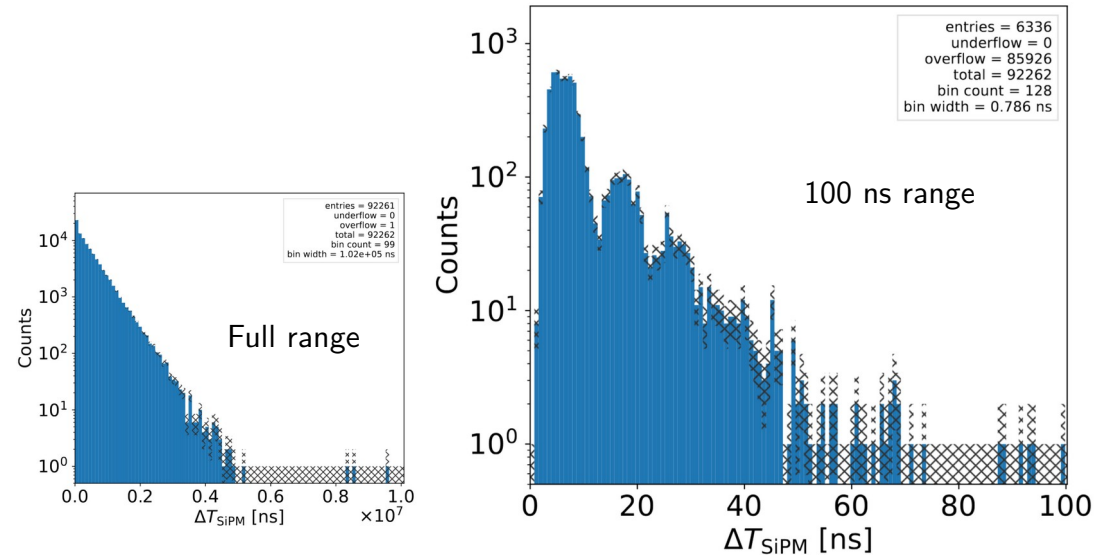


SiPM hit data

Without dead time:

- Observe systematic effects in SiPM hit data
 - High **SiPM rate** $O(1 \text{ kHz})$ for selected threshold
 - **Secondary hits** with discrete timing
- ▶ Possible signal reflections or problems with sampling logic
 - ▶ Apply dead time as simple countermeasure

Time difference ΔT_{SiPM} between hits of same SiPM:



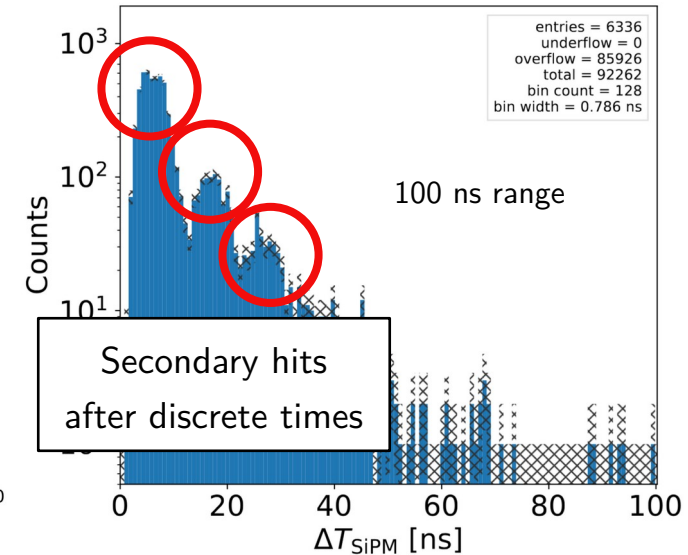
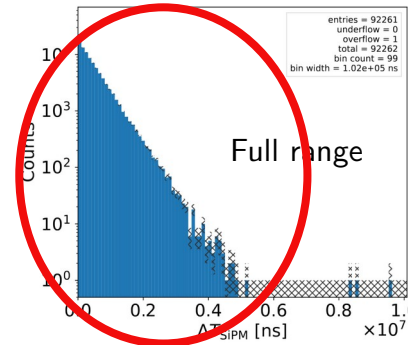
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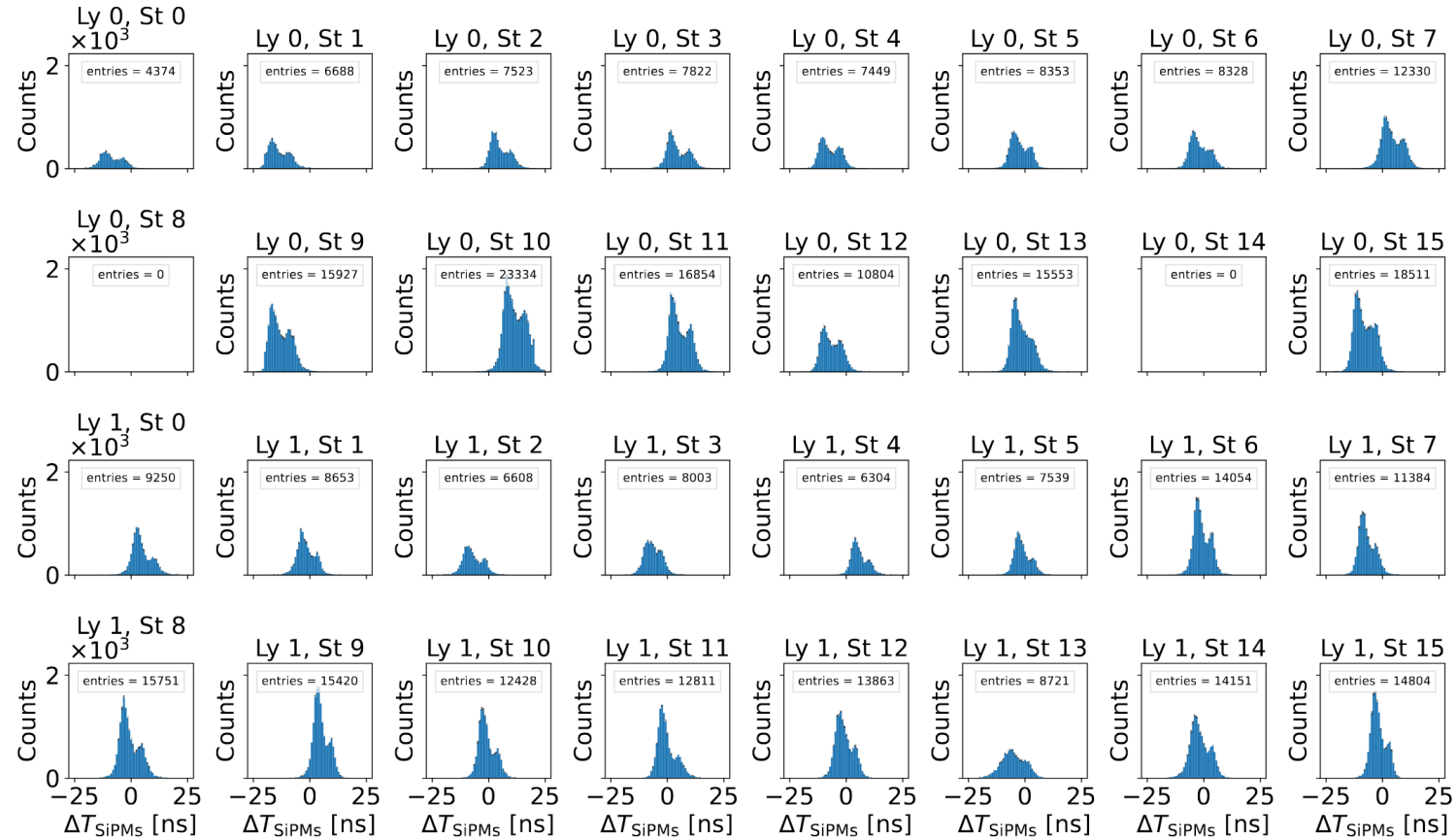
Exponential distribution:
Poisson random events



Scintillator strip coincidences

Time difference ΔT_{SiPMs} of coincidences for all strips:

- Observe double-peak structure in time difference of SiPMs
- Needs future investigation
 - In this work, accept all coincidence hits, even though structure is visible



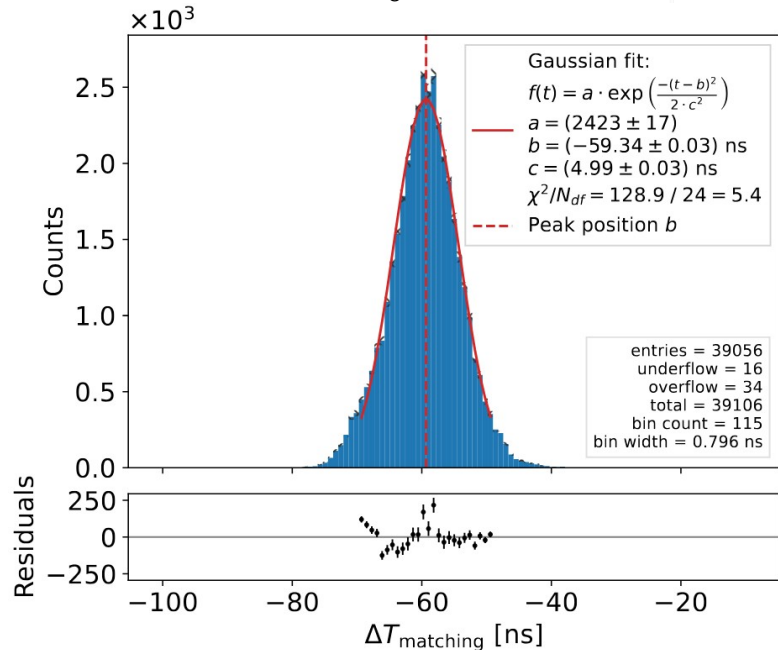
Backup (Pt. 4)

- Combined data analysis

Temporal matching of DT tracks & scintillator hits

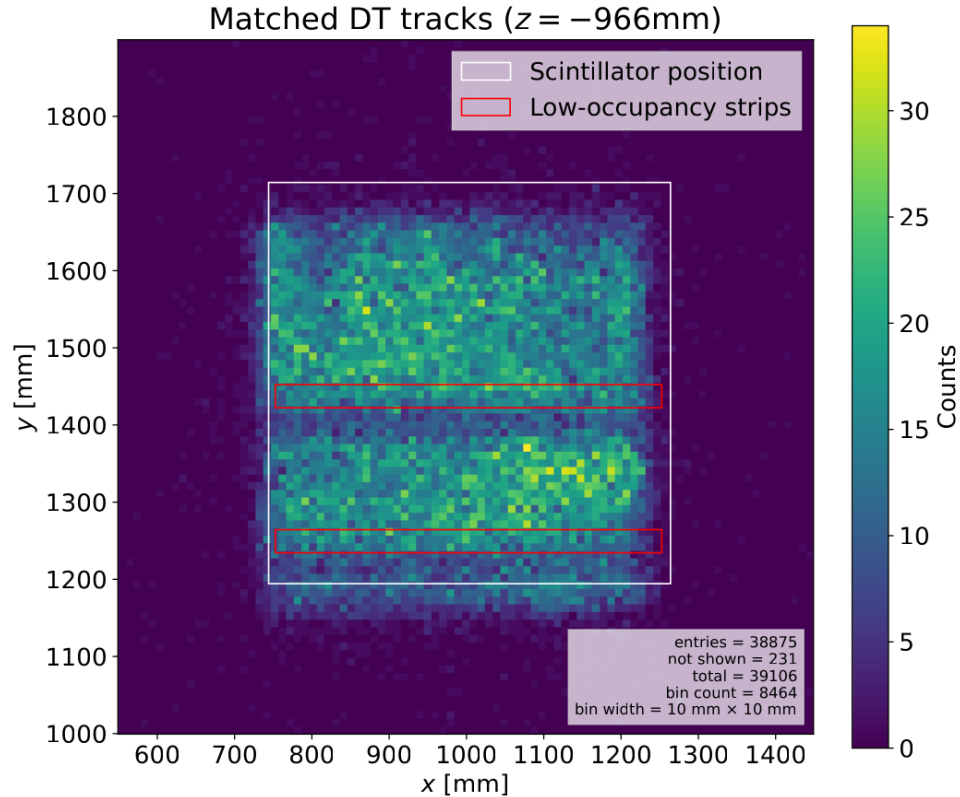
- Timing peak is not really Gaussian
 - Expected because SiPMs all have individual timing variations of up to 6 ns (uncertainty of scintillator timing calibration)
 - Rather: Sum of Gaussians of each individual scintillator strip

Time difference $\Delta T_{\text{matching}}$ of matched hits with Gaussian fit:



Independent cross-check of DT track reconstruction

2D histogram of matched tracks in scintillator:



- Fraction of matched DT tracks not hitting the scintillator is **$O(1\%)$**
 - See entries in stat-box in plot

Backup (Pt. 5)

- Simplified Monte Carlo simulation with cosmics

Monte-Carlo generation of atmospheric muon tracks

- Randomly generate atmospheric muon tracks
 - Base point (x,y): Chamber area ± 1.5 m, uniformly distributed
 - Base point z: Lowest point of chamber, fixed
 - Azimuth Φ : Uniformly distributed
 - Zenith Θ : Sample $\sim \cos^2 \Theta$ distribution
 - Propagate tracks through detector geometry
 - DT chamber: Check for cell intersection at height of respective wire, Respect cell efficiency by “dicing”
 - Scintillator: Check for strip intersection at mean height
 - Perform exactly the same reconstruction methods as for data (discussed above)
- Very simplified “geometry only” approach !
- No modeling of noise & no modeling of spatial / timing uncertainty
 - No modeling of “physics”, i.e. muon interactions and effects in detector
 - Assumption of track hit at central height of detector device
 - At wire height in drift cell (strong simplification) & mean strip height in scintillator